# Detection and Forgetting under Asymmetric Data Availability in Class-Incremental Network Intrusion Detection

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## Abstract

Class-incremental network intrusion detection must control false alarms while retaining old attack classes. We study benign data available outside attack replay memory, using a refreshed, independently calibrated benign–attack gate and an attack-only family classifier. Original AA-CIL combines strong attack detection with substantial forgetting: 20.04 and 11.12 pp on UNSW-NB15 and NSL-KDD. The validation-selected CBF variant failed the retention criterion with 15.32 pp forgetting on UNSW. AA-CIL-KD reduced forgetting to 2.31 and 0.98 pp while achieving attack-detection rates of 70.53 and 71.67%, respectively, at nominal calibration target $\alpha = 2\%$. KD forgetting was not significantly higher than the lowest-forgetting external baseline on either dataset. However, its final balanced-accuracy advantage over the matched-exposure Replay-BS control on NSL-KDD was not significant ($p = 0.1329$), and its NSL test FPR was 7.94%. The false-alarm–forgetting–accuracy trade-off therefore remains unresolved. KD was not the validation-selected main version; UNSW is exploratory and NSL is not blind confirmation. **Keywords:** continual learning; intrusion detection; benign data availability; false-positive calibration; replay memory.

## 1 Introduction

Intrusion detection models must recognize newly observed attack families without discarding knowledge of older ones. In class-incremental learning, historical attack records are represented by a limited replay memory, and inference must choose among all observed classes without a task identifier. This constraint is operationally meaningful when previous attack captures cannot be retained indefinitely or reacquired. It does not necessarily imply that benign traffic has the same availability: a monitored network may continue to provide benign examples.

The asymmetry is useful only under an explicit assumption. The learner must have access to a sufficiently clean, representative benign source, distinct from its test data. This paper does not assume that arbitrary production traffic is correctly labeled benign. Nor do we assume that a training-domain benign holdout guarantees the same false-positive rate (FPR) after a distribution change. Both qualifications matter in our results.

Replay-based NIDS research includes class balancing, frequency-aware replacement, representative selection, and other memory designs [1–4]. Some multiclass formulations place Normal and attack classes in a common classifier and replay budget. However, this is not a universal description of the literature: binary and novelty-detection methods use different protocols, and prior domain-adaptation work has already used new benign data outside an old attack memory [5]. The question is therefore not whether benign data have ever been treated differently. It is whether a continuously refreshed, independently calibrated detector provides useful retention and detection behavior in a multistage class-incremental experiment.

We study availability-asymmetric class-incremental learning (AA-CIL). A binary gate $G$ is updated using newly sampled benign data, current attacks, and stored old attacks. A separate classifier $F$ predicts attack families. A benign-only calibration pool sets $G$'s operating threshold without test labels. Memory allocation to attack families is evaluated as a secondary component rather than presented as the principal mechanism.

The contribution is an evidence-bounded experimental study:

- An explicit separation of persistent attack replay, refreshed benign training data, and permanent benign calibration data, with each storage and training cost disclosed.
- A modular gate/family design and matched-protocol comparisons that distinguish restricted benign refresh from attack-memory reallocation.
- A low-false-alarm evaluation reporting actual test FPR, binary attack detection, family-level performance, and negative registered outcomes rather than relying on aggregate accuracy alone.
- A separately identified post-hoc control, Replay-BS, demonstrating a dataset-dependent difference under matched benign exposure while retaining the limitation of unmatched total computation.

The study follows the original superiority protocol, the validation-selected retention protocol, and the fixed-KD reassessment, followed by a fixed-capacity retention extension. Their criteria and chronology are given in Appendix A; each retains its own outcome.

# 2 Related Work

## 2.1 Replay and class-incremental bias

iCaRL combines exemplar retention with incremental representation learning [6]. Dark Experience Replay and DER++ preserve historical responses alongside supervised rehearsal [7]. These approaches motivate strong replay references, but results obtained with image architectures and original buffering schemes are not automatically reproduced by a tabular NIDS adaptation.

Several methods address old/new-class imbalance. BiC learns a post-training correction [8]; weight alignment adjusts classifier scale [9]; separated softmax and taskwise distillation reduce cross-task interference [10]; end-to-end incremental learning includes balanced fine-tuning [11]. Logit adjustment uses class-prior information to change decision scores [12]. We use these methods to contextualize earlier diagnostic experiments, not as evidence that a binary gate is a new idea. Adaptation details and their unsuccessful results are retained rather than generalized to the original algorithms.

## 2.2 Continual intrusion detection

CBRS-NIDS distinguishes within-attack and across-class balancing [1]. ECBRS uses frequency-driven replacement and minority protection, while PAPA addresses replay interference [2]. SSF selects and forgets examples according to their distributional relevance [3]. These are predefined *rules*, not necessarily numerically fixed class quotas. SPIDER combines semi-supervised learning and gradient-projection memory, retaining previous-task unlabeled information rather than implementing an all-history equal-class labeled buffer [13]. CND-IDS studies continual novelty detection with latent regularization and a benign reference set [14]. ACIL evaluates multiple memory strategies and reports class-level behavior [4]. Binary domain-incremental, novelty-detection, and multiclass class-incremental results must not be treated as interchangeable benchmarks.

Our targeted source review found that most of the inspected primary result tables did not separately report benign FPR at a specified operating point. ACIL provides partial or indirect benign-class reporting. This observation motivates explicit operating-point reporting here; it is not a field-wide prevalence estimate or a claim that prior work ignores false alarms.

## 2.3 The nearest availability and multi-stage precedents

Cerasuolo et al.'s attack-adaptive NIDS uses a multiclass incremental model with exemplars from old benign and attack classes [15]. Their adaptable and explainable NIDS also considers new-domain benign data; it is therefore incorrect to describe its benign information as restricted exclusively to old memory [16].

The closest availability precedent is their network-agnostic study [5]. Its adaptation experiment supplies new benign data outside an old memory containing only attacks. This defeats a broad claim that external benign data combined with attack-only old memory is unprecedented. In the inspected experiment, binary detection is derived from multiclass attack outputs, and the evaluation does not establish an independently trained gate that is repeatedly refreshed and FPR-calibrated across multiple attack increments. AACIL examines this narrower combination. The absence of an identical verified implementation is not proof of novelty, and a useful incremental contribution must still be isolated experimentally.

Wang and Cao's GLOBECOM paper describes a two-stage class-incremental approach [17]. Only its official abstract and publication record were available for this study. Its stages concern anti-forgetting learning and unbiased prediction; we cannot infer a binary-gate/family architecture, benign refresh policy, or replay exclusion rule from its title. The unavailable full text remains an explicit literature limitation. Finally, TS-mem below is our controlled restricted-refresh ablation, *not* a faithful reproduction of any of these papers.

# 3 Problem Analysis

## 3.1 A corrected experimental basis

The motivating experiments use a taskwise minimum-update schedule rather than only a small fixed number of epochs. This prevents late, rare attack tasks from receiving arbitrarily few optimizer updates. Old attack classes are accessible only through stored training and validation rows after their introduction. Frozen preparation and duplicate/conflict handling are retained across methods; "deduplicated UNSW-NB15" does not imply that every benchmark is free of feature collisions or label conflicts. Historical experiments that revisited complete old attack pools are not used as evidence for the present access-constrained protocol. That old-pool resampling was a bug in this project's earlier implementation. The undertraining and data-quality issues identified in that development history are not attributed to published methods without source-specific evidence.

## 3.2 False alarms are important, but not a universal root cause

An earlier decomposition of the historical S2000 DER++ final confusion matrices showed substantial Normal-origin false positives in CIC-IDS-2017 attack-family columns, including families with high recall but low precision. These DER++ proportions must not be assigned to every method. UNSW-NB15 also exhibited attack-to-attack confusions and lost attack recall. These observations justify measuring benign FPR and attack detection separately, but do not identify inadequate benign coverage or Normal–attack memory competition as a universal causal explanation. The decomposition is arithmetic, not an intervention proving a mechanism.

Under the registered correction sweep, the best

tested correction/base combinations closed only 4.92% and 19.23% of the positive Joint-minus-incremental Macro-F1 gaps on UNSW-NB15 and CIC-IDS-2017, respectively. This rejects the sufficiency of those specific adaptations and candidate ranges, not bias correction in general. Joint is an offline reference rather than a guaranteed numerical upper bound.

### 3.3 Why change the learning organization?

The preceding FARQ experiments adjusted shared replay allocation using measured validation false alarms. Their registered main criterion failed, and descriptive operating curves did not establish a universally superior false-alarm/detection frontier. Replay+LA and Replay share the same attainable predictions when the total adjustment is recalibrated; this does not make all allocation methods threshold-equivalent. WACB and Freq change the learned model and improve detection relative to Replay at evaluated matched-FPR points on UNSW-NB15 and CIC-IDS-2017. Memory reallocation can therefore improve the measured trade-off, although it does not close every offline-reference gap. These finite operating-point comparisons are not a global frontier theorem. AA-CIL tests a different learning organization, not a premise that every shared-classifier method merely moves along one common curve.

## 4 Method: AA-CIL

### 4.1 Data roles and access assumption

At update $t$, let $\mathcal{A}_t$ be the observed attack families and $\mathcal{N}_t$ the newly introduced attack families. The learner receives their current training data $D_t^A$, retained old attack training memory $R_{t-1}^{A,\mathrm{tr}}$, and a benign training source $B^{\mathrm{tr}}$. A permanent benign calibration pool $B^{\mathrm{cal}}$ is disjoint from every optimizer input. We assume continued access to $B^{\mathrm{tr}}$ but not to the original pools of old attacks.

The benchmark simulates benign availability by drawing a task-dependent sample $B_t \subseteq B^{\mathrm{tr}}$ of up to 50000 rows. Samples may overlap across tasks; this is repeated access to a fixed prepared training source, not evidence from a newly collected live stream. The permanent split prevents a calibration row from later becoming a training row.

### 4.2 Independent detector and family learner

The gate is a binary MLP trained on

$$D_t^G = B_t \cup D_t^A \cup R_{t-1}^{A,\mathrm{tr}}. \tag{1}$$

Its labels are benign versus attack. The family MLP is trained on

$$D_t^F = D_t^A \cup R_{t-1}^{A,\mathrm{tr}}, \tag{2}$$

using attack-family labels only. Both use the existing class-balanced cross-entropy implementation and retain learned parameters between updates. Their optimizers and costs are separate. Neither model receives calibration data as optimization input.

For exact implementation fidelity, the family network is initialized with the benchmark's full attack-label output dimension. Training follows the shared replay cross-entropy implementation, while inference restricts the argmax to observed attack families. Thus this experiment assumes a known finite benchmark label universe; it is not a claim about an unbounded, dynamically expanding unknown-label vocabulary.

### 4.3 Attack replay allocation

Let $K_t = |\mathcal{A}_t| + 1$, including Normal, and let $m$ be the common per-observed-class budget. The shared capacity is

$$M_t = mK_t, \qquad m = 200. \tag{3}$$

AA-CIL divides $M_t$ among observed attacks: $q_{t,c} = \lfloor M_t/|\mathcal{A}_t| \rfloor$ plus one for the first remainder classes in ascending class-ID order. This is a capacity ceiling, not a promise of equal actual occupancy. Each class allocation contains separate training and validation slots; validation rows do not enter optimization. When a class is new, examples can be selected from its prepared training and validation partitions. Once it is old, the corresponding retained arrays are the only permissible source, even if a later quota exceeds available rows. Unused slots remain unused; old pools are never revisited to fill them.

This allocation transfers the nominal Normal share to attacks. It does not make benign data free in a physical-storage sense: both benign training samples and the permanent calibration pool are disclosed outside $M_t$. The no-transfer ablation tests whether this allocation adds value beyond benign refresh.

### 4.4 Calibration and inference

Let $s_G(x) = z_A(x) - z_B(x)$ be the gate's attack-minus-benign logit margin. For $n = |B^{\mathrm{cal}}|$, sort its margins as $s_{(1)} \le \cdots \le s_{(n)}$ and set

$$k_\alpha = \lfloor \alpha n \rfloor, \qquad q_{t,\alpha} = s_{(n-k_\alpha)}. \tag{4}$$

The strict decision $s_G(x) > q_{t,\alpha}$ bounds the empirical calibration false-alarm count by $k_\alpha$, including ties. No randomized tie breaking is used. Prediction is

$$\hat{y}(x) = \begin{cases} \text{Normal}, & s_G(x) \le q_{t,\alpha}, \\ \arg\max_{c \in \mathcal{A}_t} F_c(x), & s_G(x) > q_{t,\alpha}. \end{cases} \tag{5}$$

This is an empirical training-holdout operating point, not a distribution-free guarantee on test or deployment FPR. Gate false negatives cannot be recovered by the downstream family classifier.

### 4.5 Post-hoc retention extension: AA-CIL-KD

The original AA-CIL gate relies on attack replay without an explicit response-retention loss, and its family

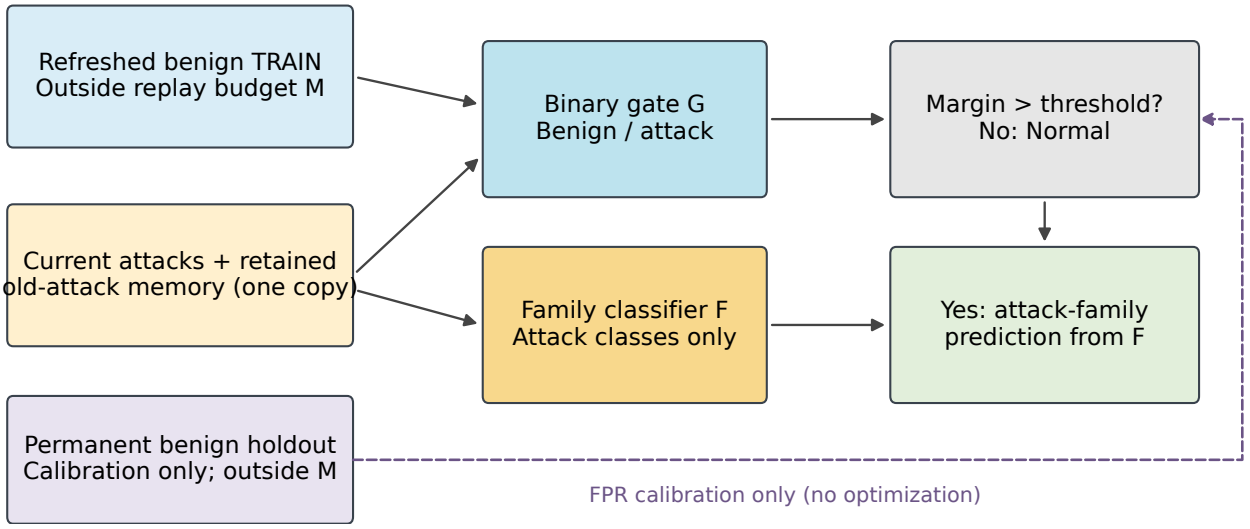



Figure 1: AA-CIL separates detection from family classification. The calibration pool never enters training. A single retained attack memory supplies both learners and is counted once; two networks and both training schedules are counted in cost.

**Algorithm 1** One AA-CIL update

**Require:** Current attacks $D_t^A$; retained attack memory $R_{t-1}^A$; benign pools $B^{\text{tr}}, B^{\text{cal}}$

1: Sample $B_t$ from $B^{\text{tr}}$ only.
2: Train $G$ on $B_t \cup D_t^A \cup R_{t-1}^{A,\text{tr}}$.
3: Train $F$ on $D_t^A \cup R_{t-1}^{A,\text{tr}}$.
4: Allocate attack capacities $q_{t,c}$ within $M_t$.
5: **for** each observed attack $c$ **do**
6: Select new-class rows, or subsample retained old rows; keep training and validation disjoint.
7: **end for**
8: Calibrate $q_{t,\alpha}$ on $B^{\text{cal}}$; do not update weights.
9: Save models, retained memory, allocation, costs, and operating-point confusion matrices.

learner uses ordinary replay. AA-CIL-KD is a post-hoc extension to that study, with its protocol and success criteria registered before observing any KD results. It adds a gate loss

$$\mathcal{L}_G = \mathcal{L}_{\text{CE},G} + \lambda_G \mathcal{L}_{\text{MSE},G}. \tag{6}$$

For retained old attack training rows, the gate targets are the stored two-logit responses recorded when those rows entered memory. For the current benign sample, targets come from the frozen preceding-task gate. New attack rows have no old-class distillation target, and the first task has no preceding-task gate. Neither calibration rows nor attack validation rows become optimizer inputs.

The attack-only family learner uses a task-boundary DER++-objective adaptation: supervised learning is supplemented by replay cross-entropy and MSE against stored family logits. This is not a claim to reproduce the original DER++ online storage schedule. Gate and family logits are auxiliary records for the same attack rows, not two separately allocated sample memories; their storage and the two training costs must be disclosed. The update budgets, benign refresh, permanent holdout and threshold policy remain those of AA-CIL.

An initial development search used mean stagewise validation balanced accuracy and varied three distillation coefficients. It was superseded before formal KD results by the retention-constrained selection below. Its completed runs remain part of the development cost, but do not enter the new selection. The previously planned noGKD ablation is not one of the five family-model candidates in this revised experiment.

### 4.6 Post-hoc parameter isolation: AA-CIL-ISO

AA-CIL-ISO changes only the family learner relative to AA-CIL-KD: the gate, G-KD objective, benign refresh and calibration remain the same. It uses the existing MLP feature encoder. On the first task containing attacks, the encoder and that task's new family heads train jointly within one family-model update budget. The encoder is then frozen. Each new family $c$ receives its own rank-8 residual adapter and two-logit binary head,

$$h_c = h + U_c V_c h, \qquad p_c(x) = \text{softmax}(H_c h_c)_1. \tag{7}$$

The head learns that family versus the other attacks available in the current task and legal old-attack replay. Multiple new families provide negatives for one another. Normal rows are not used as family-head negatives. After the task, its adapters and heads are frozen permanently; old attack pools are never reopened.

All heads introduced in one task share a single training loop: their class-balanced binary cross-entropies are averaged before each optimizer step. Each new head participates in the task's common

max(8 epochs, 2000 steps) schedule, rather than receiving an additional serial budget per head. The first encoder training has no separate pretraining budget. Joint family-loop time and shared steps are recorded once; exclusive per-head time is not inferred from that total. If the first task contains only one attack family, there are no other-attack negatives: this limitation is recorded explicitly, without invented negatives or benign substitution.

After the gate declares an attack, prediction is $\arg\max_{c \in \mathcal{A}_t} p_c(x)$ over independently normalized positive probabilities. This is head-only inference with no router and no cross-head softmax calibration. It adapts OFRA-style parameter isolation to the present backbone and is evaluated under the validation-selected retention protocol, with descriptive paired comparisons against KD.

### 4.7 Retention alternatives and validation-only selection

The revised experiment compares five family learners under the same G-KD gate: KD, ISO, weight alignment (WA), class-balanced fine-tuning (CBF), and nearest-class-mean prediction (NCM). WA rescales newly introduced classifier weights by the mean old-to-new weight-norm ratio after task training [9]. CBF adds small-learning-rate updates on balanced retained attack training memory [11]. NCM computes class prototypes from retained attack training rows encoded by the current family encoder and predicts by feature-space distance [6]. These are adaptations of established mechanisms, not claims of novel versions of the cited algorithms. Their additional time, steps, prototype storage and model storage are reported separately.

Only UNSW-NB15 development seeds $\{5, 6\}$ select hyperparameters and the main version. Each candidate uses $\lambda_G \in \{0.5, 1, 2\}$; KD fixes both family DER coefficients to one. CBF uses 0.1 times the base learning rate and either one or five fine-tuning epochs. A stratified permanent 10% attack holdout supplies fixed validation cohorts across stages; it is never optimized, is accessible only for this development evaluation, and its storage is disclosed outside the replay budget. The permanent Normal holdout supplies the gate threshold at $\alpha = 2\%$ and the Normal validation cohort. This development-only access exception does not reopen old attack training pools or change formal replay access.

For each version, eligible configurations have final validation balanced accuracy at least the original AA-CIL development mean minus three percentage points. The configuration with lowest mean validation forgetting is retained, and the lowest-forgetting eligible version is selected as the candidate main method. All choices are frozen across datasets before evaluation seeds $\{1, 2, 3, 4, 42\}$ are used. Every version is reported, not only the selected one. If no configuration is eligible, there is no eligible main method; any configuration retained for descriptive formal reporting is explicitly a fallback, not an eligibility success. Test labels never select the gate threshold, hyperparameters, or main version.

### 4.8 Fixed-capacity retention candidates

A subsequent extension constrains the family learner to a fixed parameter count across tasks, no more than twice the full original family model, at most one extra component, and a single inference model. KD-weak reduces the family DER coefficients; KD+WA and KD+CBF apply alignment or balanced fine-tuning after KD. EMA-KD maintains one slow family copy (decay 0.999), uses replay-logit MSE with weight one, and predicts only with that copy. DUAL uses two fixed encoders and a joint concatenation head: the stable encoder has the frozen KD coefficient, while the plastic encoder has no distillation loss. The all-attack output capacity is allocated initially; unseen classes remain masked until introduction. There is no taskwise branch expansion.

The fixed-capacity retention protocol uses both UNSW-NB15 and NSL-KDD development seeds $\{5, 6\}$ with the fixed attack validation cohorts and permanent Normal holdout. It requires each dataset's validation forgetting to be at most its lowest external-five value plus one percentage point before comparing mean validation final BAcc. Within one point of BAcc, the simpler candidate is preferred. Gate coefficients are $\lambda_G \in \{0.25, 0.5\}$; KD-weak searches each DER coefficient in $\{0.25, 0.5\}$, and the other KD configurations retain coefficients of one. CBF uses one epoch at 0.1 times the base learning rate. If no candidate meets both forgetting constraints, the lowest-mean-forgetting version is frozen only as an explicitly ineligible descriptive fallback. All five candidates are reported on formal seeds, with UNSW and NSL still designated development datasets.

## 5 Experimental Setup

### 5.1 Benchmarks and task protocol

We use the prepared NSL-KDD, deduplicated UNSW-NB15, and CIC-IDS-2017 benchmarks [18–20]. Normal and an initial attack family appear in the first task, followed by the fixed attack-family sequence. The data manifest and generated protocol table specify label mappings, task groupings, split sizes, and feature dimensions. Test rows from unobserved classes are excluded at intermediate checkpoints; final metrics use the complete prepared test set. No test row is selected for training, memory construction, or primary calibration.

Training uses MLPs with two hidden layers of width 128, Adam with learning rate 0.001, and batch size 256. Each model/update receives

$$S_t = \max(2000, 8 \lceil |D_t|/256 \rceil) \tag{8}$$

optimizer steps. Consequently, matching this schedule does not imply identical realized steps when data exposure differs. Gate and family schedules are recorded separately. The registered seed set contains 5 seeds;

Table 1: Dataset/task and final-budget summary. Calibration bytes include float32 features and int64 row indices, exclude model state and dense labels. The pool is shared by all methods, excluded from optimization, and outside M.

| Dataset | Classes | Tasks | M ceiling | Calib rows | Calib bytes |
|---|---|---|---|---|---|
| NSL-KDD | 5 | 4 | 1000 | 6735 | 3178920 |
| UNSW-NB15 | 10 | 5 | 2000 | 6825 | 1911000 |
| CIC-IDS-2017 | 8 | 4 | 1600 | 181176 | 57251616 |

Table 2: Prepared split sizes, feature counts and fixed task groupings (class IDs). These are the exact prepared benchmark arrays, not an assertion of temporal deployment validity.

| Dataset | TRAIN rows | TEST rows | Features | Task IDs (semicolon separates tasks) |
|---|---|---|---|---|
| NSL-KDD | 125973 | 22544 | 116 | 0,1; 2; 3; 4 |
| UNSW-NB15 | 120991 | 30248 | 68 | 0,1; 2,3; 4,5; 6,7; 8,9 |
| CIC-IDS-2017 | 2262203 | 565673 | 77 | 0,1; 2,3; 4,5; 6,7 |

NSL-KDD: 0=Normal, 1=DoS, 2=Probe, 3=R2L, 4=U2R. UNSW-NB15: 0=Normal, 1=Generic, 2=Exploits, 3=Fuzzers, 4=DoS, 5=Reconnaissance, 6=Analysis, 7=Backdoor, 8=Shellcode, 9=Worms. CIC-IDS-2017: 0=Normal, 1=DoS, 2=DDoS, 3=Bruteforce, 4=PortScan, 5=WebAttack, 6=Botnet, 7=Infiltration.

tables report actual completed $n$ where the cutoff prevents a complete group. Partial groups are not silently replaced with selected favorable seeds.

The observed local GPU is an NVIDIA GeForce RTX 4060 Ti with 8 GiB memory; the HPC execution environment uses NVIDIA A100 GPUs. HPC results are included only after completed execution, not merely submission or allocation. Heterogeneous hardware and shared local GPU use mean that reported runtimes are not a controlled hardware benchmark.

## 5.2 Comparators and ablations

The external comparison set is Replay, DER++, WACB, Freq, and Replay+LA. Replay uses the common tabular architecture. DER++ is the recorded tabular adaptation with classwise memory management, not an exact replication of the original vision protocol. WACB adapts benign-versus-combined-attack balancing to this multiclass setting. Freq is a simplified frequency-proportional allocation control, not an exact ECBRS reproduction. Replay+LA is a saved-model logit-adjustment evaluation; it is not an additional independently trained network. All primary operating thresholds are determined from the same permanent benign calibration pool.

TS-mem retains the same gate/family organization but restricts each task's benign refresh to the training portion of the ordinary Normal allocation, while sharing replay capacity between Normal and attacks. It therefore measures a restricted benign exposure policy, not the effect of a frozen benign subset alone. AA-noAttMem retains refreshed benign training and the independent gate but does not transfer the Normal allocation to attacks. Its name does not mean that attacks have no replay memory.

Joint trains with the offline data available under its recorded protocol and is reported as a reference. It is not a guaranteed upper bound, a matched-storage online learner, or a member of the strongest-baseline selection. FARQ is supplementary. The registered selection includes the external methods *and* TS-mem and AA-noAttMem; excluding those ablations after inspecting results would change the registered question.

## 5.3 Metrics and registered inference

For confusion counts $C_{ij}$ with Normal index $b$, we report

$$\mathrm{FPR} = \frac{\sum_{j \neq b} C_{bj}}{\sum_j C_{bj}}, \tag{9}$$

$$\mathrm{ADR} = \frac{\sum_{i \neq b} \sum_{j \neq b} C_{ij}}{\sum_{i \neq b} \sum_j C_{ij}}. \tag{10}$$

ADR is binary attack detection, not correct-family recall. Macro-F1 and attack-family macro recall supplement it. The registered family-recall guard includes attack classes meeting the specified minimum test support; excluded rare classes remain visible in per-class evidence. Precision is undefined for a class with no predictions in a seed and is never changed to zero. Its group mean and sample standard deviation use only defined seed values, with the actual $n_{\text{defined}}$ (`defined_n`) reported separately from the total seed count. If none are defined, the mean remains undefined; with fewer than two defined values, the sample standard deviation is undefined. Such a seed is not dropped from recall, forgetting, or other defined endpoints.

The primary calibration target is $\alpha = 2\%$. Actual test FPR is reported alongside it because equal calibration targets need not yield equal test FPR. The original superiority protocol requires a one-sided paired test against the highest-mean registered comparator in at least two datasets. The comparator is chosen once per dataset and endpoint from whole-seed means, not separately per seed. Other criteria concern Macro-F1 harm, attack-family recall, the CIC gate-refresh ablation, and descriptive cost/Joint reporting. Paired differences, sample standard deviations and effect sizes accompany complete five-seed comparisons. Appendix A defines all three protocols and their distinct comparator sets.

# 6 Results

## 6.1 Class-incremental retention and forgetting

Class-incremental performance is evaluated before the false-alarm–detection comparison. For every saved checkpoint $t = 1, \ldots, T$, the operating threshold is determined anew from the same permanent Normal calibration pool at $\alpha = 2\%$, using that checkpoint's scores. The final threshold is not retrospectively applied to earlier models. For a seen class $c$, let $r_{c,t}$ denote its recall under this operating point and let $\mathcal{C}_t$ be the seen-class set. We report

$$B_t = \frac{1}{|\mathcal{C}_t|} \sum_{c \in \mathcal{C}_t} r_{c,t}, \qquad \overline{B}_{\text{inc}} = \frac{1}{T} \sum_{t=1}^{T} B_t, \tag{11}$$

together with final balanced accuracy $B_T$. Mean incremental balanced accuracy weights stages equally, rather than pooling their examples or reporting micro accuracy.

Let $\tau_c$ be the introduction stage of class $c$ and $\mathcal{O} = \{c : \tau_c < T\}$. Signed final forgetting is

$$\mathcal{F} = \frac{1}{|\mathcal{O}|} \sum_{c \in \mathcal{O}} \left( \max_{\tau_c \le t < T} r_{c,t} - r_{c,T} \right). \tag{12}$$

The historical maximum excludes the final stage; negative values therefore retain final improvement and are not clipped to zero. The primary definition includes Normal whenever it belongs to $\mathcal{O}$; an attack-only version excludes Normal from this same set. Classes introduced in the final task have no subsequent update and are not assigned a forgetting value. No rare-class support filter is applied to these Class-IL summaries: the support filter belongs only to the separate attack-family safety guard. We report per-seed values and their mean and sample standard deviation, with forgetting in percentage points. Low forgetting alone is not a ranking of useful retention: a class that was never learned well has little recall to lose. It must be read together with final and mean incremental balanced accuracy for every method. Incomplete trajectories or seed groups remain explicitly incomplete.

The completed original-AA trajectories show why detection and retention must be separated. On UNSW-NB15, mean incremental BAcc is 69.65%, final BAcc is 55.05%, and signed forgetting is 20.04±1.16 percentage points. On NSL-KDD, the corresponding values are 74.24%, 58.26%, and 11.12±2.70 points. A higher trajectory average therefore coexists with substantial old-class recall loss; the new KD and ISO variants target that unresolved endpoint rather than converting the earlier detection gain into a retention claim.

For each old attack class, the confusion matrix partitions its error into a gate miss (prediction Normal) and a wrong attack-family prediction. At the same class-specific pre-final peak used above, the recall loss equals the increase in these two error rates. Both components use the same class support and peak checkpoint.

The validation-selected retention protocol makes retention mandatory on both UNSW-NB15 and NSL-KDD. Its selected version must satisfy the forgetting, final-BAcc and ADR conditions in Appendix A.2. CIC-IDS-2017 is supplementary to this protocol. Mean incremental BAcc, Macro-F1 and actual test FPR accompany all five variants.

UNSW training-validation seeds 5/6 locked AA-CIL-CBF ($\lambda_G = 2$, 1 CBF epoch(s), learning-rate multiplier 0.1) as the eligible main version. AA-CIL-KD, AA-CIL-ISO, AA-CIL-NCM are reported as development-ineligible fallback configurations. Complete formal five-seed groups: UNSW 5/5 versions, NSL 5/5 versions. On UNSW-NB15, AA-CIL-ISO recorded signed forgetting of 0.018±0.032 pp alongside final balanced accuracy of 19.67±0.02%. ISO minus KD differences were -2.29±1.03 pp for forgetting ($p_< = 0.00385$) and -18.86±1.10 pp for mean incremental balanced accuracy ($p_< = 1.37 \times 10^{-6}$), using five paired seeds and unadjusted one-sided tests. The validation-selected retention protocol fails a necessary condition; the false-alarm–forgetting–accuracy trade-off remains unresolved.

**Separately scoped KD reassessment.** The fixed-KD reassessment uses $\lambda_G = .5$ and DER $\alpha = \beta = 1$. On NSL-KDD, the final BAcc difference from Replay-BS is 1.49±2.58 pp ($p_> = 0.1329$); the required superiority condition is not met. A necessary numerical condition fails; the trade-off remains unresolved. Across all five CIC17 KD seeds, mean incremental BAcc is 87.59±1.07%, final BAcc is 79.20±4.08%, and signed forgetting is -10.54±9.15 pp. At the nominal 2% calibration target, test FPR, ADR, and Macro-F1 are 1.93±0.04%, 92.10±2.84%, and 65.94±2.49%, respectively.The CIC-IDS-2017 KD group is complete (5/5 seeds). Paired comparisons requiring Replay-BS (4/5 seeds) remain pending.

**Fixed-capacity retention extension.** The fixed-capacity retention protocol completed validation-only selection at 2026-10-20 03:17:14.796356+08:00. None of the five candidates satisfies the forgetting constraint on both development datasets. KD+WA is retained only as an ineligible descriptive fallback with the lowest average signed validation forgetting, with $\lambda_G = 0.5$ and family DER coefficients $\alpha = 1$, $\beta = 1$. This selection did not meet the retention requirement and does not resolve the trade-off. 9 of ten five-seed candidate/dataset groups are complete. UNSW and NSL remain development datasets. CSE-CIC-IDS2018 metrics are sealed and were not read by this presentation; no external confirmation or resolved-trade-off conclusion is available.

## 6.2 Primary calibration-set operating points

Table 13 reports methods at the common calibration target, actual test FPR and completed seed counts at 2026-10-20 04:44 MYT.

Table 3: NSL: checkpoint-calibrated Class-IL performance at $\alpha = 2\%$.

| Method | $n$ | Mean inc. BAcc | Final BAcc | Forget (all old) | Forget (old att.) | Final ADR | Final MF1 |
|---|---|---|---|---|---|---|---|
| Replay | 5 | 70.12±2.82 | 47.05±4.52 | 9.73±4.30 | 12.53±5.85 | 65.39±5.30 | 46.83±4.93 |
| Replay+LA | 5 | 70.12±2.82 | 47.05±4.52 | 9.73±4.30 | 12.53±5.85 | 65.39±5.30 | 46.83±4.93 |
| DER++ | 5 | 68.13±2.81 | 50.11±2.97 | 0.91±5.03 | 1.04±6.67 | 65.28±2.73 | 50.46±3.15 |
| WACB | 5 | 73.98±2.05 | 56.69±5.22 | 4.16±2.86 | 5.43±3.67 | 66.83±3.77 | 57.03±3.91 |
| Freq | 5 | 74.82±2.70 | 61.82±2.44 | 5.25±2.76 | 6.35±3.65 | 69.67±2.53 | 56.84±4.32 |
| AA-CIL | 5 | 74.24±1.29 | 58.26±1.85 | 11.12±2.70 | 14.75±3.70 | 62.84±3.46 | 55.66±3.36 |
| TS-mem | 5 | 72.65±2.03 | 55.62±1.58 | 10.53±3.63 | 13.73±4.66 | 63.19±4.66 | 51.44±2.16 |
| AA-noAttMem | 5 | 74.31±1.46 | 56.97±2.69 | 13.78±3.47 | 18.82±4.61 | 60.29±3.16 | 56.67±3.34 |
| Replay-BS (post hoc) | 5 | 75.05±1.01 | 59.32±2.85 | 11.44±2.53 | 15.15±3.51 | 62.48±4.64 | 57.05±3.33 |
| AA-CIL-KD | 5 | 76.39±0.26 | 60.82±0.73 | 0.98±0.76 | 1.27±1.17 | 71.67±0.66 | 61.43±2.25 |
| AA-CIL-ISO | 5 | 58.29±0.60 | 36.50±0.58 | 0.06±0.14 | -0.06±0.05 | 72.41±0.24 | 31.77±1.09 |
| AA-CIL-WA | 5 | 76.17±0.67 | 59.33±1.09 | 2.37±0.59 | 3.02±0.66 | 72.41±0.24 | 59.85±1.87 |
| AA-CIL-CBF | 5 | 76.01±0.62 | 59.00±1.27 | 2.34±0.50 | 2.98±0.81 | 72.41±0.24 | 59.78±2.59 |
| AA-CIL-NCM | 5 | 76.16±0.27 | 59.98±0.68 | 2.35±1.00 | 3.09±1.33 | 71.67±0.66 | 60.05±1.81 |
| FARQ (supp.) | 5 | 74.57±2.57 | 58.22±4.10 | 4.70±5.00 | 5.99±6.40 | 67.45±2.77 | 58.86±5.79 |
| FARQ-all (supp.) | 5 | 73.21±3.72 | 54.47±12.14 | 14.21±10.57 | 18.86±14.08 | 63.08±2.48 | 49.57±11.97 |
| Joint (offline) | 5 | 74.62±0.69 | 58.65±2.20 | 3.45±0.81 | 4.51±0.93 | 66.63±1.80 | 59.37±2.90 |

Mean±sample SD across the available seeds; accuracies are percentages and forgetting is in percentage points. Mean incremental BAcc averages all seen-class stage BAcc values. Signed forgetting compares each old class's best pre-final recall with its final recall; the primary scope includes Normal. Smaller forgetting alone is not superiority: inspect final and incremental BAcc together. Groups with $n < 5$ are incomplete, descriptive only. Joint is offline; FARQ is supplementary; Replay+LA shares Replay's calibrated total $\tau$ and incurs no separate inference. The validation-selected retention protocol reports all five F variants; the main variant is fixed by development seeds 5/6, not by this test table. These post-hoc stagewise summaries do not change the original superiority protocol decision.

Table 4: UNSW: checkpoint-calibrated Class-IL performance at $\alpha = 2\%$.

| Method | $n$ | Mean inc. BAcc | Final BAcc | Forget (all old) | Forget (old att.) | Final ADR | Final MF1 |
|---|---|---|---|---|---|---|---|
| Replay | 5 | 52.29±0.55 | 25.15±0.89 | 5.09±1.02 | 5.82±1.17 | 38.52±3.77 | 26.27±0.85 |
| Replay+LA | 5 | 52.29±0.55 | 25.15±0.89 | 5.09±1.02 | 5.82±1.17 | 38.52±3.77 | 26.27±0.85 |
| DER++ | 5 | 38.73±1.38 | 14.43±1.03 | 16.16±1.80 | 18.44±2.04 | 18.48±4.12 | 14.47±1.44 |
| WACB | 5 | 55.12±1.53 | 29.96±1.30 | 3.48±2.45 | 3.97±2.81 | 48.97±4.14 | 32.81±1.18 |
| Freq | 5 | 55.83±1.60 | 28.93±1.40 | 5.50±1.25 | 6.27±1.43 | 50.42±2.93 | 31.22±1.73 |
| AA-CIL | 5 | 69.65±0.49 | 55.05±1.06 | 20.04±1.16 | 22.92±1.32 | 69.09±0.93 | 41.18±0.28 |
| TS-mem | 5 | 48.14±3.67 | 24.97±7.34 | 9.73±5.85 | 11.11±6.69 | 24.79±14.93 | 23.47±6.33 |
| AA-noAttMem | 5 | 69.28±0.39 | 53.98±1.02 | 22.02±0.87 | 25.16±0.99 | 67.94±1.58 | 40.43±0.51 |
| Replay-BS (post hoc) | 5 | 62.16±0.24 | 38.01±1.19 | 4.87±2.44 | 5.54±2.78 | 66.14±0.74 | 39.99±1.57 |
| AA-CIL-KD | 5 | 63.75±1.11 | 41.80±1.93 | 2.31±1.07 | 2.62±1.20 | 70.53±0.50 | 42.08±1.93 |
| AA-CIL-ISO | 5 | 44.89±0.03 | 19.67±0.02 | 0.02±0.03 | 0.01±0.02 | 71.01±1.16 | 11.47±0.01 |
| AA-CIL-WA | 5 | 68.25±0.47 | 54.24±0.79 | 16.21±1.09 | 18.51±1.25 | 71.01±1.16 | 41.35±0.32 |
| AA-CIL-CBF | 5 | 68.73±0.49 | 54.69±0.70 | 15.32±0.56 | 17.49±0.64 | 71.01±1.16 | 40.86±0.38 |
| AA-CIL-NCM | 5 | 68.18±0.16 | 51.32±1.10 | 18.12±2.76 | 20.69±3.15 | 70.53±0.50 | 38.22±1.08 |
| FARQ (supp.) | 5 | 54.99±1.32 | 27.90±1.46 | 6.43±2.31 | 7.35±2.63 | 45.10±4.20 | 30.25±1.68 |
| FARQ-all (supp.) | 5 | 50.90±2.38 | 23.24±3.19 | 8.41±3.96 | 9.58±4.51 | 34.56±10.80 | 24.30±3.56 |
| Joint (offline) | 5 | 62.06±0.52 | 35.38±0.67 | 5.35±1.10 | 6.10±1.26 | 67.11±1.45 | 37.22±1.00 |

Mean±sample SD across the available seeds; accuracies are percentages and forgetting is in percentage points. Mean incremental BAcc averages all seen-class stage BAcc values. Signed forgetting compares each old class's best pre-final recall with its final recall; the primary scope includes Normal. Smaller forgetting alone is not superiority: inspect final and incremental BAcc together. Groups with $n < 5$ are incomplete, descriptive only. Joint is offline; FARQ is supplementary; Replay+LA shares Replay's calibrated total $\tau$ and incurs no separate inference. The validation-selected retention protocol reports all five F variants; the main variant is fixed by development seeds 5/6, not by this test table. These post-hoc stagewise summaries do not change the original superiority protocol decision.

On UNSW-NB15, AA-CIL obtains ADR 69.09%, Macro-F1 41.18%, and test FPR 1.92%. Relative to the strongest *external* ADR baseline, the gain is 18.67 percentage points. This comparison is descriptive with an associated paired test, distinct from the registered strongest-comparator conclusion.

AA-CIL's UNSW-NB15 Worms recall is 63.75%, with precision 3.52%; Fuzzers recall is 9.67%. The appendix reports the corresponding difficult-family metrics.

On NSL-KDD, AA-CIL reaches ADR 62.84%, compared with 69.67% for Freq. Its Macro-F1 is 55.66%, compared with 57.03% for WACB. Its test FPR, 6.15%, is substantially different from the calibration target. The method therefore neither wins across benchmarks nor delivers the nominal low-FPR operating point on this shifted test distribution.

At $\alpha = 2\%$, CIC-IDS-2017 AA-CIL has all five seeds: actual test benign FPR 1.93±0.02%, attack detection 83.66±2.83%, and Macro-F1 63.15±1.14% (mean ± sample SD). Against DER++ from the registered seven-comparator set, AA-CIL minus comparator ADR is -5.63±4.15 pp (one-sided $p = 0.98070$), which does not meet the dataset-level positive-difference and one-sided $p < 0.05$ condition. Against DER++ from the external-only comparator set, AA-CIL minus comparator ADR is -5.63±4.15 pp (one-sided $p = 0.98070$), which does not meet the dataset-level positive-difference and one-sided $p < 0.05$ condition.

## 6.3 Outcome of the original superiority protocol

On UNSW-NB15, the strongest registered ADR comparator is AA-noAttMem, not Freq. AA-CIL's paired mean advantage is 1.15 percentage points, with one-sided $p = 0.0748$. It does not pass the registered significance criterion. NSL-KDD also fails the superiority requirement. Even a favorable CIC-IDS-2017 result cannot then provide superiority in the required majority of datasets. The primary registered criterion and conjunction are therefore negative; remaining individ-

Table 5: CIC17: checkpoint-calibrated Class-IL performance at $\alpha = 2\%$.

| Method | $n$ | Mean inc. BAcc | Final BAcc | Forget (all old) | Forget (old att.) | Final ADR | Final MF1 |
|---|---|---|---|---|---|---|---|
| Replay | 5 | 74.30±1.20 | 52.32±4.55 | 6.99±7.30 | 8.38±8.76 | 78.89±5.63 | 51.52±4.02 |
| Replay+LA | 5 | 74.30±1.20 | 52.32±4.55 | 6.99±7.30 | 8.38±8.76 | 78.89±5.63 | 51.52±4.02 |
| DER++ | 5 | 76.85±0.97 | 48.44±2.90 | 17.52±6.03 | 21.02±7.23 | 89.29±3.01 | 48.85±3.19 |
| WACB | 5 | 77.43±1.19 | 63.02±1.31 | 1.97±3.48 | 2.36±4.17 | 88.91±2.30 | 57.84±1.13 |
| Freq | 5 | 76.38±1.80 | 64.03±2.62 | 4.71±9.31 | 5.63±11.16 | 88.55±2.32 | 57.97±4.33 |
| AA-CIL | 5 | 93.95±0.46 | 87.54±0.85 | 8.99±1.57 | 10.77±1.88 | 83.66±2.83 | 63.15±1.14 |
| TS-mem | 5 | 66.04±1.55 | 40.34±2.01 | 26.70±3.69 | 32.03±4.42 | 49.51±3.90 | 34.63±2.25 |
| AA-noAttMem | 5 | 93.00±0.39 | 87.50±0.35 | 9.49±0.90 | 11.38±1.07 | 83.13±1.48 | 63.60±0.78 |
| Replay-BS (post hoc) | 4 | 91.17±1.90 | 79.39±4.16 | 5.35±10.84 | 6.41±13.01 | 92.62±1.24 | 69.03±1.05 |
| AA-CIL-KD | 5 | 87.59±1.07 | 79.20±4.08 | -10.54±9.15 | -12.66±10.99 | 92.10±2.84 | 65.94±2.49 |
| FARQ (supp.) | 5 | 75.72±1.79 | 57.54±5.40 | 1.20±7.71 | 1.43±9.25 | 85.40±5.48 | 55.13±2.26 |
| FARQ-all (supp.) | 5 | 71.58±3.86 | 50.30±9.58 | 8.28±15.62 | 9.93±18.73 | 74.49±10.03 | 48.47±6.84 |
| Joint (offline) | 5 | 93.83±0.80 | 78.68±3.54 | 13.14±2.71 | 15.75±3.26 | 98.19±1.94 | 68.43±1.30 |

Mean±sample SD across the available seeds; accuracies are percentages and forgetting is in percentage points. Mean incremental BAcc averages all seen-class stage BAcc values. Signed forgetting compares each old class's best pre-final recall with its final recall; the primary scope includes Normal. Smaller forgetting alone is not superiority: inspect final and incremental BAcc together. Groups with $n < 5$ are incomplete, descriptive only. Joint is offline; FARQ is supplementary; Replay+LA shares Replay's calibrated total $\tau$ and incurs no separate inference. The validation-selected retention protocol reports all five F variants; the main variant is fixed by development seeds 5/6, not by this test table. These post-hoc stagewise summaries do not change the original superiority protocol decision.

Table 6: NSL: AA-CIL old-attack recall-loss decomposition at $\alpha = 2\%$.

| Old attack | $n$ | Recall loss (pp) | Δ G miss (pp) | Δ F wrong (pp) |
|---|---|---|---|---|
| DoS | 5 | 14.71±3.92 | 13.96±4.01 | 0.75±0.34 |
| Probe | 5 | 25.06±5.05 | 25.40±5.60 | -0.34±1.31 |
| R2L | 5 | 4.49±9.23 | -5.77±13.47 | 10.27±5.49 |

For each seed and class, the reference is its earliest highest-recall stage before the final task. Normal-miss and wrong-attack-family rates use all true-class rows as denominator; their signed changes add to recall loss. Means add, while component SDs do not. This observed confusion-matrix decomposition does not infer F behavior on G-rejected rows.

ual criteria retain their actual complete or incomplete states.

The separately reported external-baseline comparison does not alter this protocol's strongest-comparator definition or negative outcome.

## 6.4 What the ablations identify

On UNSW-NB15, TS-mem achieves ADR 24.79%, compared with 67.94% for AA-noAttMem. The latter uses greater benign refresh while retaining ordinary attack allocations.

The AA-CIL–AA-noAttMem contrast measures attack-memory transfer. Its UNSW-NB15 detection difference is small relative to the external-baseline gap and is not significant in the registered one-sided test.

## 6.5 Post-hoc exposure-matched and shift analyses

Replay-BS was added after the initial results as an exposure-matched control. Its single multiclass replay model uses the same refreshed benign samples, attack-only memory capacity, current attacks and calibration policy as AA-CIL.

On UNSW-NB15, AA-CIL improves attack detection over Replay-BS by 2.95 percentage points in the post-hoc paired test ($p = 0.0029$; Table 15). On NSL-KDD, the difference is 0.35 points ($p = 0.4052$), with higher test FPR and lower Macro-F1 than Replay-BS. The exposure-matched detection result therefore differs across datasets.

The diagnostic NSL sensitivity analysis contrasts training-domain benign-holdout thresholds with thresholds estimated using test-domain Normal labels. Test-Normal calibration brings the measured false-alarm rate to the target by construction but reduces attack detection and Macro-F1 for the evaluated gate variants (Table 17).

The post-hoc operating curves in Figs. 2–4 show the sampled false-alarm–detection and Macro-F1 trade-offs. Finite-grid points without a bracket remain unavailable; no extrapolation is applied.

## 6.6 Cost and storage

We charge the gate and family model separately and report their sum. Selection, calibration, evaluation and additional benign sample storage are separate quantities. Realized replay occupancy is reported alongside capacity because rare old classes may leave slots unused.

The permanent benign pool and the extra benign training inputs of AA-CIL, AA-noAttMem and Replay-BS are outside the replay ceiling but inside the disclosed total data footprint. TS-mem instead uses its retained Normal training rows within the ceiling. Normal validation rows already counted within TS-mem's replay allocation are not counted again as additional unique calibration rows. Successful optimization time excludes loading and serialization unless explicitly stated. Local GPU sharing and heterogeneous execution devices preclude interpreting raw runtime differences as an exclusive hardware benchmark. Failed, duplicate, or superseded executions are not scientific replicates and are accounted for separately from selected successful-cell costs.

Table 7: UNSW: AA-CIL old-attack recall-loss decomposition at $\alpha = 2\%$.

| Old attack | $n$ | Recall loss (pp) | Δ G miss (pp) | Δ F wrong (pp) |
|---|---|---|---|---|
| Generic | 5 | 14.34±0.84 | 0.89±0.97 | 13.45±1.49 |
| Exploits | 5 | 43.78±3.40 | 4.31±1.91 | 39.46±5.06 |
| Fuzzers | 5 | 21.64±1.42 | 15.61±1.27 | 6.03±1.10 |
| DoS | 5 | 28.83±4.10 | 3.92±4.01 | 24.91±2.80 |
| Reconnaissance | 5 | 24.57±9.16 | 6.19±2.98 | 18.38±7.19 |
| Analysis | 5 | 25.56±8.35 | 26.67±7.93 | -1.11±1.11 |
| Backdoor | 5 | 1.71±8.17 | -8.29±6.88 | 10.00±3.64 |

For each seed and class, the reference is its earliest highest-recall stage before the final task. Normal-miss and wrong-attack-family rates use all true-class rows as denominator; their signed changes add to recall loss. Means add, while component SDs do not. This observed confusion-matrix decomposition does not infer F behavior on G-rejected rows.

Table 8: CIC17: AA-CIL old-attack recall-loss decomposition at $\alpha = 2\%$.

| Old attack | $n$ | Recall loss (pp) | Δ G miss (pp) | Δ F wrong (pp) |
|---|---|---|---|---|
| DoS | 5 | 19.22±0.63 | 18.31±0.98 | 0.91±0.57 |
| DDoS | 5 | 3.70±1.33 | 3.56±1.36 | 0.14±0.18 |
| Bruteforce | 5 | 3.34±2.80 | 3.09±2.79 | 0.24±0.24 |
| PortScan | 5 | 25.30±8.44 | 25.10±8.37 | 0.21±0.15 |
| WebAttack | 5 | 2.30±2.94 | 0.22±0.16 | 2.08±2.92 |

For each seed and class, the reference is its earliest highest-recall stage before the final task. Normal-miss and wrong-attack-family rates use all true-class rows as denominator; their signed changes add to recall loss. Means add, while component SDs do not. This observed confusion-matrix decomposition does not infer F behavior on G-rejected rows.

# 7 Discussion and Limitations

## 7.1 A useful result without a universal winner

The strongest supported observation is dataset dependent: independently calibrated benign-refresh models substantially improve the selected external-baseline comparison on UNSW-NB15, whereas NSL-KDD does not improve the relevant registered endpoints. The evidence does not support a blanket superiority claim. In particular, favorable binary ADR can coexist with poor family discrimination: detected attacks assigned to the wrong family count as detected in ADR but not as correct in the multiclass metrics.

The work also separates a design hypothesis from its identification. TS-mem shows the outcome of restricting benign refresh within a two-model architecture. The completed Replay-BS comparison strengthens the UNSW-NB15 evidence beyond an explanation based solely on additional benign exposure. Its NSL-KDD result does not support the same benefit, and its single network still receives less total model training than the gate-plus-family configuration. The supported claim is therefore an exposure-matched, dataset-dependent configuration benefit, not a compute-isolated causal estimate of decomposition.

The TS-mem–AA-noAttMem contrast changes benign exposure, realized gate updates and temporary storage together. The attack-memory contrast isolates a different allocation decision within that design, but its nonsignificant detection difference does not establish zero benefit or dispensability. Similarly, the gate/family decomposition partitions observed errors; it cannot by itself attribute forgetting to weight interference, because the operating threshold also changes between checkpoints.

The TS-mem comparison gives paired AA-CIL minus comparator differences of 34.14±5.78 pp in ADR and 28.52±1.74 pp in Macro-F1; the registered one-sided Macro-F1 test has $p = 1.67\times10^{-6}$. The AA-noAttMem comparison gives paired AA-CIL minus comparator differences of 0.53±4.02 pp in ADR and -0.45±1.76 pp in Macro-F1 (one-sided ADR $p = 0.39082$). The post-hoc Replay-BS comparison is incomplete (AA-CIL $n = 5/5$, Replay-BS (post hoc) $n = 4/5$); no complete-group inference is made. Regardless of this additional comparison, the failed original superiority requirement is unchanged. A CIC-only gain would not establish a universal advantage, and a post-hoc curve or Replay-BS contrast would remain distinct from that registered decision.

## 7.2 Calibration and deployment assumptions

Clean labeled benign data are a substantial resource. They may be contaminated, costly to curate, or distributionally unrepresentative of future traffic. Refresh from a fixed training pool is only a benchmark surrogate for continuous benign availability. It does not validate resilience to poisoned benign pools, temporal drift, network migration, or delayed labels. NSL-KDD's calibration-to-test shift illustrates why a nominal calibration target must not be equated with realized deployment FPR.

The test-Normal sensitivity analysis deliberately uses test labels and is not a deployable calibration rule or a replacement for the primary result. Threshold tightening exchanges false alarms for missed attacks, and same-population calibration does not establish out-of-sample reliability. Post-hoc mean curves at a common threshold and averages of per-seed interpolated ADR at a fixed FPR are different summaries; neither changes the holdout-calibrated protocol decisions.

Table 9: Validation-selected retention protocol: frozen development configurations.

| Version | $\lambda_G$ | CBF epochs | CBF LR multiplier | Development role |
|---|---|---|---|---|
| AA-CIL-KD | 0.5 | – | – | ineligible fallback |
| AA-CIL-ISO | 2 | – | – | ineligible fallback |
| AA-CIL-WA | 2 | – | – | eligible alternative |
| AA-CIL-CBF | 2 | 1 | 0.1 | eligible main |
| AA-CIL-NCM | 0.5 | – | – | ineligible fallback |

Selection uses UNSW training-validation seeds 5/6 only. DER weights are fixed at one. Ineligible versions are still reported on the five formal seeds, without converting fallback selection into eligibility.

Table 10: Retention variants at stagewise Normal-holdout calibration target $\alpha = 2\%$.

| Data | Version | n | Mean inc. BAcc | Final BAcc | Forget (pp) | ADR | MF1 | FPR |
|---|---|---|---|---|---|---|---|---|
| UNSW | AA-CIL-KD | 5 | 63.75±1.11 | 41.80±1.93 | 2.31±1.07 | 70.53±0.50 | 42.08±1.93 | 2.10±0.18 |
| UNSW | AA-CIL-ISO | 5 | 44.89±0.03 | 19.67±0.02 | 0.02±0.03 | 71.01±1.16 | 11.47±0.01 | 2.04±0.08 |
| UNSW | AA-CIL-WA | 5 | 68.25±0.47 | 54.24±0.79 | 16.21±1.09 | 71.01±1.16 | 41.35±0.32 | 2.04±0.08 |
| UNSW | AA-CIL-CBF | 5 | 68.73±0.49 | 54.69±0.70 | 15.32±0.56 | 71.01±1.16 | 40.86±0.38 | 2.04±0.08 |
| UNSW | AA-CIL-NCM | 5 | 68.18±0.16 | 51.32±1.10 | 18.12±2.76 | 70.53±0.50 | 38.22±1.08 | 2.10±0.18 |
| NSL | AA-CIL-KD | 5 | 76.39±0.26 | 60.82±0.73 | 0.98±0.76 | 71.67±0.66 | 61.43±2.25 | 7.94±0.45 |
| NSL | AA-CIL-ISO | 5 | 58.29±0.60 | 36.50±0.58 | 0.06±0.14 | 72.41±0.24 | 31.77±1.09 | 8.40±0.37 |
| NSL | AA-CIL-WA | 5 | 76.17±0.67 | 59.33±1.09 | 2.37±0.59 | 72.41±0.24 | 59.85±1.87 | 8.40±0.37 |
| NSL | AA-CIL-CBF | 5 | 76.01±0.62 | 59.00±1.27 | 2.34±0.50 | 72.41±0.24 | 59.78±2.59 | 8.40±0.37 |
| NSL | AA-CIL-NCM | 5 | 76.16±0.27 | 59.98±0.68 | 2.35±1.00 | 71.67±0.66 | 60.05±1.81 | 7.94±0.45 |

Only complete seeds 1/2/3/4/42 receive means and sample SD; incomplete groups show n and dashes. All outcomes come from formal test trajectories, never development selection metrics. Signed forgetting includes old Normal; rates/accuracies are percentages. Eligibility is a development property, not formal success.

### 7.3 Scope of statistical and architectural claims

The benchmark uses few paired seeds, fixed task orders, a finite known label universe, prepared tabular features, and a shared training implementation. These choices support controlled within-protocol comparisons but do not cover arbitrary task order, unseen network environments, or open-world label discovery. A matched schedule is not matched total computation or exposure. The fixed-capacity extension uses both UNSW and NSL for validation selection, so even its later five-seed results on these datasets are developmental, not blind confirmation. CSE-CIC-IDS2018 outcomes remain sealed and are not part of the present evidence. Joint remains an offline reference whose available information differs from incremental methods.

All reported paired tests are unadjusted. Selection of a comparator from the highest or lowest observed whole-group mean and the small seed count limit inferential strength. A nonsignificant harm test is not an equivalence or noninferiority result. KD was not selected as the primary version by validation: its UNSW-NB15 reassessment is exploratory, and its NSL-KDD outcomes were already available before direct authorization. The first two NSL seeds also predate the reassessment document. We therefore do not describe NSL as blind confirmation or infer whether those available results had been inspected.

The initial manuscript window left the CIC-IDS-2017 KD group incomplete. A supplemental retention job reached the scheduler's `DEADLINE` state without allocation or execution; that job produced no scientific outcome. The subsequent deadline-free continuation completed all five KD and all five original AA-CIL trajectories. Replay-BS remains at four of five seeds, with seed 42 absent, so the paired final-BAcc comparison with that control remains incomplete. Missing trajectories are missing evidence rather than negative model results, and partial-group means are descriptive only.

The targeted literature review does not establish first-of-its-kind status. In particular, external benign data and attack-only old memory already appear in neighboring adaptation work. Our restricted-refresh ablation is not an implementation of that literature. Full-text unavailability for Wang and Cao further limits its precise differentiation. A novel substantive contribution would require verified distinction and an identifiable benefit beyond combining a gate, replay, and threshold calibration; neither architecture naming nor an external-baseline gain alone meets that standard.

### 7.4 Relation to the broader research questions

This manuscript addresses predictive retention, class-wise safety, and cost under class-incremental updates. It does not measure silent explanation drift across checkpoints, and it does not evaluate non-suppressive explanation governance. Those explanatory and governance questions remain unanswered by these experiments. Predictive gains alone cannot establish stable explanations or the safety of an explanation-governance mechanism.

## 8 Conclusion

Class-incremental NIDS must balance false alarms, attack detection and retention as new families arrive. Benign-stream training with an independently calibrated gate supports strong attack detection in the studied setting, but the original decoupled model exhibits 20.04 and 11.12 pp forgetting on UNSW-NB15 and NSL-KDD. The validation-selected CBF variant failed the retention criterion; KD reduced forgetting to 2.31 and 0.98 pp with attack-detection rates of 70.53

Table 11: Fixed-capacity protocol: validation-only choices, not formal test results.

| Candidate | $\lambda_G$ | DER $\alpha/\beta$ | Mean val. BAcc | Mean val. forget | UNSW | NSL |
|---|---|---|---|---|---|---|
| KD-weak | 0.5 | 0.25/0.5 | 73.82 | 5.79 | no | no |
| KD+WA | 0.5 | 1/1 | 73.18 | 2.09 | yes | no |
| KD+CBF | 0.5 | 1/1 | 73.40 | 6.40 | no | no |
| EMA-KD | 0.5 | 1/1 | 71.11 | 4.71 | no | no |
| DUAL | 0.5 | 1/1 | 73.26 | 7.81 | no | no |

Seeds 5/6 only. BAcc is in percent and signed forgetting in percentage points; means average the two development datasets. The last two columns indicate each dataset's forgetting constraint. Ceilings are 4.68 pp (UNSW) and -17.73 pp (NSL), each equal to its lowest external-five validation forgetting plus 1 pp. Negative values are not clipped. Rows with no eligible configuration are fixed descriptive fallbacks.

Table 12: Fixed-capacity candidates on development datasets at stagewise $\alpha = 2\%$.

| Data | Candidate | $n$ | Mean inc. BAcc | Final BAcc | Forget (pp) | ADR | MF1 | FPR |
|---|---|---|---|---|---|---|---|---|
| UNSW | KD-weak | 5 | 70.08±0.49 | 56.73±1.57 | 11.75±1.22 | 70.53±0.50 | 43.39±0.69 | 2.10±0.18 |
| UNSW | KD+WA | 5 | 68.63±0.70 | 55.17±1.26 | 5.70±0.97 | 70.53±0.50 | 44.77±0.34 | 2.10±0.18 |
| UNSW | KD+CBF | 5 | 70.15±0.45 | 56.66±1.51 | 11.68±1.73 | 70.53±0.50 | 43.53±0.71 | 2.10±0.18 |
| UNSW | EMA-KD | 5 | 69.80±0.51 | 56.50±1.00 | 8.51±1.22 | 70.53±0.50 | 43.74±0.71 | 2.10±0.18 |
| UNSW | DUAL | 5 | 69.93±0.38 | 56.63±1.04 | 14.77±2.35 | 70.53±0.50 | 42.29±0.33 | 2.10±0.18 |
| NSL | KD-weak | 5 | 76.81±0.49 | 61.36±0.96 | 1.52±0.63 | 71.67±0.66 | 61.38±1.61 | 7.94±0.45 |
| NSL | KD+WA | 5 | 76.77±0.41 | 61.19±0.59 | 1.53±0.85 | 71.67±0.66 | 61.58±1.21 | 7.94±0.45 |
| NSL | KD+CBF | 5 | 76.83±0.30 | 61.30±1.35 | 1.65±1.11 | 71.67±0.66 | 60.62±1.60 | 7.94±0.45 |
| NSL | EMA-KD | 5 | 76.15±0.55 | 59.97±0.63 | 2.88±1.15 | 71.67±0.66 | 57.01±1.25 | 7.94±0.45 |

Mean±sample SD over complete seeds 1/2/3/4/42. Both UNSW and NSL were used for development; these five-seed results are not blind confirmation. Only complete groups are displayed. No test outcome changes the fixed selection or fallback eligibility.

and 71.67% at nominal $\alpha = 2\%$. Nevertheless, its final balanced-accuracy advantage over the matched-exposure Replay-BS control on NSL-KDD was not significant ($p = 0.1329$), while the NSL test FPR of 7.94% exposes calibration shift. These results support a trade-off study, not a fully resolved false-alarm–forgetting–accuracy solution. KD was not the validation-selected main version; UNSW is exploratory and NSL is not blind confirmation.

The resulting contribution is a reproducible study of this unresolved trade-off under an explicit benign-data availability assumption. Clean benign access, calibration shift, unmatched total computation and the incomplete CIC-IDS-2017 exposure-matched reference limit generalization. Silent explanation drift and non-suppressive explanation governance remain separate, unanswered research questions.

# Reproducibility and evidence availability

Tables and numerical prose are generated from repository result files rather than manually transcribed. The companion code retains training configurations, seeds, taskwise confusion matrices, sample-access policies, and separate gate/family costs. Model payloads and datasets are not embedded in the manuscript. The evidence cutoff and actual group sizes are reported; unfinished results are not inferred from a running process or an anticipated completion time.

# A Decision protocols and documented chronology

This appendix makes the decision rules self-contained. Times below are the repository's document-recording timestamps, not claims of external preregistration or evidence that a reader had inspected results. Unless stated otherwise, times are UTC+08:00 on 19 October 2026. Formal comparisons use the fixed paired seeds $\{1, 2, 3, 4, 42\}$, one-sided paired $t$ tests at $p < 0.05$, and thresholds set only from the common permanent Normal training holdout at $\alpha = 2\%$. A partial group supplies descriptive means at its actual $n$, not a five-seed significance conclusion. A known necessary-condition failure remains a failure when another comparison is incomplete.

## A.1 Original superiority protocol

The revised original criteria were recorded at 14:31:21 (revision `aa16079`); the common holdout-calibration and separate gate/family update-budget clarification was recorded at 14:34:56 (`fe42938`). The comparator set contains Replay, DER++, WACB, Freq, Replay+LA, TS-mem and AA-noAttMem. The strongest comparator is chosen once per dataset and endpoint from the highest complete five-seed mean, never independently for individual seeds. The conjunction requires:

1. ADR significantly greater than the strongest comparator in at least two of NSL-KDD, UNSW-NB15 and CIC-IDS-2017.

2. Macro-F1 not significantly lower than its strongest comparator on any dataset; a one-sided lower-tail $p < 0.05$ identifies harm.

3. Attack-family macro recall not significantly lower than Replay, using attack classes with at least 20 test examples. All other per-class results remain reported.

4. On CIC-IDS-2017, AA-CIL Macro-F1 significantly greater than TS-mem Macro-F1.

Table 13: Final-task test performance at a benign-holdout calibration target of $\alpha = 2\%$. Rates are percentages; mean ± sample SD. Actual test FPR is not fixed at 2%.

| Data | Method | n | Test FPR | Attack detection | Macro-F1 | BAcc | Attack guard |
|---|---|---|---|---|---|---|---|
| NSL | Replay | 5 | 4.08±2.16 | 65.39±5.30 | 46.83±4.93 | 47.05±4.52 | 34.83±5.78 |
| NSL | DER++ | 5 | 2.51±0.45 | 65.28±2.73 | 50.46±3.15 | 50.11±2.97 | 38.27±3.77 |
| NSL | WACB | 5 | 3.62±0.87 | 66.83±3.77 | 57.03±3.91 | 56.69±5.22 | 46.77±6.49 |
| NSL | Freq | 5 | 4.93±2.42 | 69.67±2.53 | 56.84±4.32 | 61.82±2.44 | 53.50±3.50 |
| NSL | Replay+LA | 5 | 4.08±2.16 | 65.39±5.30 | 46.83±4.93 | 47.05±4.52 | 34.83±5.78 |
| NSL | TS-mem | 5 | 4.41±2.12 | 63.19±4.66 | 51.44±2.16 | 55.62±1.58 | 45.62±1.92 |
| NSL | AA-noAttMem | 5 | 4.32±1.82 | 60.29±3.16 | 56.67±3.34 | 56.97±2.69 | 47.29±3.05 |
| NSL | AA-CIL | 5 | 6.15±1.79 | 62.84±3.46 | 55.66±3.36 | 58.26±1.85 | 49.36±1.98 |
| NSL | Replay-BS (post hoc) | 5 | 3.65±2.16 | 62.48±4.64 | 57.05±3.33 | 59.32±2.85 | 50.07±3.92 |
| NSL | Joint (offline) | 5 | 6.79±1.00 | 66.63±1.80 | 59.37±2.90 | 58.65±2.20 | 50.01±2.75 |
| UNSW | Replay | 5 | 1.78±0.20 | 38.52±3.77 | 26.27±0.85 | 25.15±0.89 | 17.04±0.98 |
| UNSW | DER++ | 5 | 1.98±0.27 | 18.48±4.12 | 14.47±1.44 | 14.43±1.03 | 5.15±1.16 |
| UNSW | WACB | 5 | 1.93±0.24 | 48.97±4.14 | 32.81±1.18 | 29.96±1.30 | 22.39±1.46 |
| UNSW | Freq | 5 | 2.01±0.26 | 50.42±2.93 | 31.22±1.73 | 28.93±1.40 | 21.25±1.54 |
| UNSW | Replay+LA | 5 | 1.78±0.20 | 38.52±3.77 | 26.27±0.85 | 25.15±0.89 | 17.04±0.98 |
| UNSW | TS-mem | 5 | 1.98±0.12 | 24.79±14.93 | 23.47±6.33 | 24.97±7.34 | 16.85±8.15 |
| UNSW | AA-noAttMem | 5 | 1.94±0.19 | 67.94±1.58 | 40.43±0.51 | 53.98±1.02 | 49.09±1.12 |
| UNSW | AA-CIL | 5 | 1.92±0.15 | 69.09±0.93 | 41.18±0.28 | 55.05±1.06 | 50.27±1.18 |
| UNSW | Replay-BS (post hoc) | 5 | 2.13±0.17 | 66.14±0.74 | 39.99±1.57 | 38.01±1.19 | 31.36±1.31 |
| UNSW | Joint (offline) | 5 | 1.93±0.15 | 67.11±1.45 | 37.22±1.00 | 35.38±0.67 | 28.42±0.76 |
| CIC17 | Replay | 5 | 1.95±0.06 | 78.89±5.63 | 51.52±4.02 | 52.32±4.55 | 47.87±6.55 |
| CIC17 | DER++ | 5 | 1.92±0.07 | 89.29±3.01 | 48.85±3.19 | 48.44±2.90 | 47.68±3.63 |
| CIC17 | WACB | 5 | 1.92±0.06 | 88.91±2.30 | 57.84±1.13 | 63.02±1.31 | 59.90±0.67 |
| CIC17 | Freq | 5 | 1.93±0.02 | 88.55±2.32 | 57.97±4.33 | 64.03±2.62 | 61.81±5.97 |
| CIC17 | Replay+LA | 5 | 1.95±0.06 | 78.89±5.63 | 51.52±4.02 | 52.32±4.55 | 47.87±6.55 |
| CIC17 | TS-mem | 5 | 1.89±0.06 | 49.51±3.90 | 34.63±2.25 | 40.34±2.01 | 26.88±2.99 |
| CIC17 | AA-noAttMem | 5 | 1.94±0.03 | 83.13±1.48 | 63.60±0.78 | 87.50±0.35 | 89.21±0.46 |
| CIC17 | AA-CIL | 5 | 1.93±0.02 | 83.66±2.83 | 63.15±1.14 | 87.54±0.85 | 89.27±1.13 |
| CIC17 | Replay-BS (post hoc) | 4 | 1.94±0.03 | 92.62±1.24 | 69.03±1.05 | 79.39±4.16 | 82.56±5.86 |
| CIC17 | Joint (offline) | 5 | 1.93±0.04 | 98.19±1.94 | 68.43±1.30 | 78.68±3.54 | 78.56±3.58 |

Replay+LA and Replay share the same calibrated total logit adjustment and weights. Joint has offline old-training access. Replay-BS is post hoc. A dash means unavailable; $n < 5$ is descriptive only. The guard alone excludes attack classes with test support below 20.
CIC-IDS-2017 AA-family coverage: 15/15 cells across AA-CIL ($n = 5/5$), TS-mem ($n = 5/5$), and AA-noAttMem ($n = 5/5$); Replay-BS is reported separately. AA-CIL and AA-noAttMem refresh benign training rows outside replay memory; ordinary replay controls use quota-limited Normal training memory. Replay-BS matches the added benign exposure and attack allocation. Computational and causal limitations are discussed in Section 7.

Table 14: Original registered superiority test and separately identified external-only comparison. The latter does not replace the former.

| Data | Comparison set | Fixed mean winner | Δ ADR (pp) | $p_>$ | $d_z$ | Dataset test |
|---|---|---|---|---|---|---|
| NSL | registered | Freq | -6.829±3.350 | 0.99482 | -2.039 | Fail |
| NSL | external | Freq | -6.829±3.350 | 0.99482 | -2.039 | Fail |
| UNSW | registered | AA-noAttMem | 1.154±1.450 | 0.07480 | 0.796 | Fail |
| UNSW | external | Freq | 18.671±3.277 | 0.00011 | 5.697 | Pass |
| CIC17 | registered | DER++ | -5.632±4.149 | 0.98070 | -1.357 | Fail |
| CIC17 | external | DER++ | -5.632±4.149 | 0.98070 | -1.357 | Fail |

Original superiority protocol, criteria 1–5: 1: FAIL, 2: PASS, 3: PASS, 4: PASS, 5: PASS. Overall decision: FAIL, because NSL and UNSW cannot contribute the required two significant wins. The comparator set includes TS-mem and AA-noAttMem.

5. Descriptive reporting of the offline Joint gap and training cost, without a numerical success threshold.

This protocol's superiority condition failed and is not replaced by the later external-only or retention comparisons.

## A.2 Validation-selected retention protocol

The retention-constrained candidate-selection and decision rule was recorded at 18:42:22 (`e494c64`). Development uses only UNSW-NB15 seeds $\{5, 6\}$, permanent attack validation rows and the Normal calibration pool. Among configurations with final validation BAcc at least the original-AA development mean minus 3 percentage points, selection minimizes mean signed forgetting; this selects CBF. KD, ISO, WA, CBF and NCM are all reported on the formal seeds, including versions retained only as descriptive fallbacks after failing eligibility. The selected version must satisfy all three conditions on *both* UNSW-NB15 and NSL-KDD:

1. Signed all-old-class forgetting not significantly higher than the lowest-forgetting member of the external five: Replay, DER++, WACB, Freq and Replay+LA. The comparator minimizes the complete five-seed mean; an upper-tail $p < 0.05$ identifies harm.
2. Final BAcc significantly higher than exposure-matched Replay-BS (upper-tail $p < 0.05$).
3. ADR at the common calibration target not significantly lower than original AA-CIL (lower-tail $p < 0.05$ identifies harm).

Mean incremental BAcc, Macro-F1 and actual test FPR are mandatory descriptive outcomes. CIC-IDS-2017 is supplementary and cannot replace either required dataset. Failure of any necessary condition

Table 15: Ablations at $\alpha = 2\%$. Paired changes are AA-CIL minus the named ablation. Tests involving Replay-BS are post hoc; small groups do not amend registered decisions.

| Data | Method | n | Detection (%) | MF1 (%) | Δ detection (pp) | $p_>$ |
|---|---|---|---|---|---|---|
| NSL | AA-CIL | 5 | 62.84±3.46 | 55.66±3.36 | – | – |
| NSL | TS-mem | 5 | 63.19±4.66 | 51.44±2.16 | -0.35±4.01 | 0.5733 |
| NSL | AA-noAttMem | 5 | 60.29±3.16 | 56.67±3.34 | 2.54±3.42 | 0.0860 |
| NSL | Replay-BS (post hoc) | 5 | 62.48±4.64 | 57.05±3.33 | 0.35±3.09 | 0.4052 |
| UNSW | AA-CIL | 5 | 69.09±0.93 | 41.18±0.28 | – | – |
| UNSW | TS-mem | 5 | 24.79±14.93 | 23.47±6.33 | 44.30±15.74 | 0.0016 |
| UNSW | AA-noAttMem | 5 | 67.94±1.58 | 40.43±0.51 | 1.15±1.45 | 0.0748 |
| UNSW | Replay-BS (post hoc) | 5 | 66.14±0.74 | 39.99±1.57 | 2.95±1.23 | 0.0029 |
| CIC17 | AA-CIL | 5 | 83.66±2.83 | 63.15±1.14 | – | – |
| CIC17 | TS-mem | 5 | 49.51±3.90 | 34.63±2.25 | 34.14±5.78 | 0.0001 |
| CIC17 | AA-noAttMem | 5 | 83.13±1.48 | 63.60±0.78 | 0.53±4.02 | 0.3908 |
| CIC17 | Replay-BS (post hoc) | 4 | 92.62±1.24 | 69.03±1.05 | -8.46±2.92 | 0.9949 |

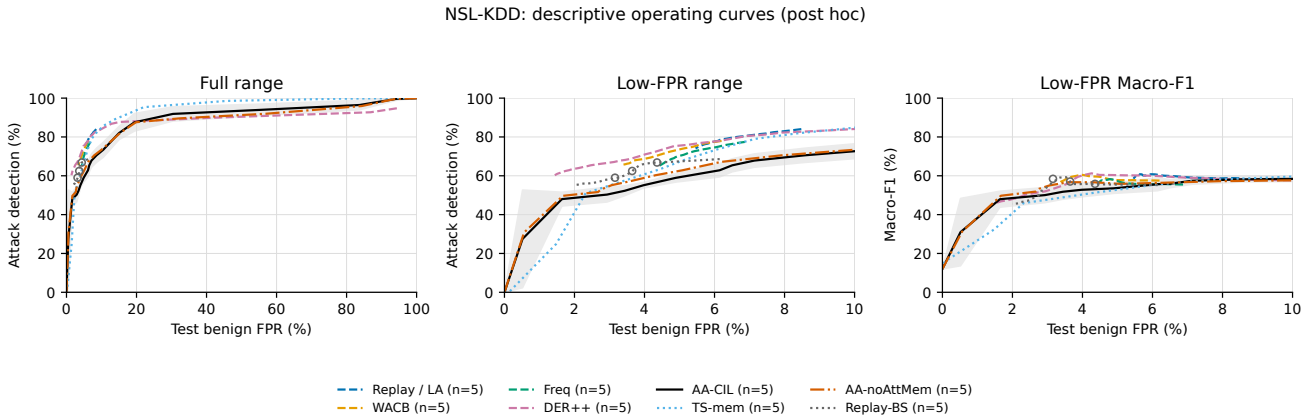



Figure 2: NSL-KDD operating curves (post hoc). Threshold-sweep summaries are descriptive, not calibration-set operating points or the registered test. Missing methods or seeds are identified in the generated panel.

Table 16: Post-hoc Replay-BS: single multiclass model with the AA-CIL benign exposure and attack replay allocation. This control was added after the original primary result.

| Data | n | Test FPR | Detection | MF1 |
|---|---|---|---|---|
| NSL | 5 | 3.65±2.16 | 62.48±4.64 | 57.05±3.33 |
| UNSW | 5 | 2.13±0.17 | 66.14±0.74 | 39.99±1.57 |
| CIC17 | 4 | 1.94±0.03 | 92.62±1.24 | 69.03±1.05 |

leaves the trade-off unresolved. The validation-selected CBF failure remains the outcome of this protocol.

## A.3 Fixed-KD reassessment

The fixed-KD reassessment was recorded at 20:47:57 (`88d6355`), after the UNSW-NB15 formal outcomes had motivated reconsidering KD. It retains the frozen $\lambda_G = 0.5$ and family DER coefficients $\alpha = \beta = 1$, with no further tuning. It applies the same three forgetting, final-BAcc and ADR conditions immediately above, but requires *both NSL-KDD and CIC-IDS-2017*; UNSW-NB15 remains exploratory. Each comparison still requires its complete five-seed candidate and reference groups. Neither this reassessment nor any favorable descriptive KD outcome changes the validation-selected CBF decision.

NSL-KDD seeds 1 and 2 had completed before the reassessment document was recorded, and all five had completed before direct execution authorization at approximately 21:29. Result availability does not establish that the results were inspected, but it rules out describing this NSL evidence as blind confirmation. The reassessment's NSL final-BAcc superiority condition failed. The remaining CIC-IDS-2017 comparison is reported at its actual completeness and cannot reverse that necessary-condition failure. The original manuscript cutoff was later removed for completion of the remaining experiments; it does not redefine the criteria or turn partial trajectories into complete results.

## A.4 Fixed-capacity retention protocol

The fixed-capacity extension was specified on 20 October 2026 before its development results. It preserves the earlier failed decisions and freezes each candidate using only UNSW/NSL validation seeds $\{5, 6\}$. Each candidate first requires signed validation forgetting no greater than the external-five minimum plus one percentage point on both datasets; among eligible configurations, mean validation final BAcc selects the configuration and main version. A BAcc tie within one point favors KD-weak/KD+WA/KD+CBF, then EMA-KD, then DUAL. An ineligible fallback is identified explicitly rather than recast as a successful selection.

The external CSE-CIC-IDS2018 assessment is separate from the development-dataset reports. Its frozen candidate must satisfy all four conditions: forgetting not significantly higher than the external-five minimum; final BAcc significantly higher than Replay-BS;

Table 17: Post-hoc NSL-KDD calibration-distribution sensitivity at $\alpha = 2\%$. Test metrics are mean ± sample standard deviation in percent.

| Method | Threshold calibration pool | n | Test FPR | ADR | MF1 |
|---|---|---|---|---|---|
| AA-CIL | TRAIN Normal holdout | 5 | 6.15±1.79 | 62.84±3.46 | 55.66±3.36 |
| AA-CIL | TEST Normal (post hoc) | 5 | 2.00±0.00 | 50.95±4.16 | 50.26±6.71 |
| TS-mem | TRAIN Normal holdout | 5 | 4.41±2.12 | 63.19±4.66 | 51.44±2.16 |
| TS-mem | TEST Normal (post hoc) | 5 | 2.00±0.00 | 53.55±3.31 | 47.81±1.00 |
| AA-noAttMem | TRAIN Normal holdout | 5 | 4.32±1.82 | 60.29±3.16 | 56.67±3.34 |
| AA-noAttMem | TEST Normal (post hoc) | 5 | 2.00±0.00 | 50.48±2.42 | 50.85±3.00 |

Each method uses the same paired seeds in both rows. TEST-Normal thresholds use test Normal labels and are evaluated on the same test population: this is a test-label-conditioned, non-deployable sensitivity analysis. The original superiority decision is unchanged; incomplete seed groups remain descriptive.

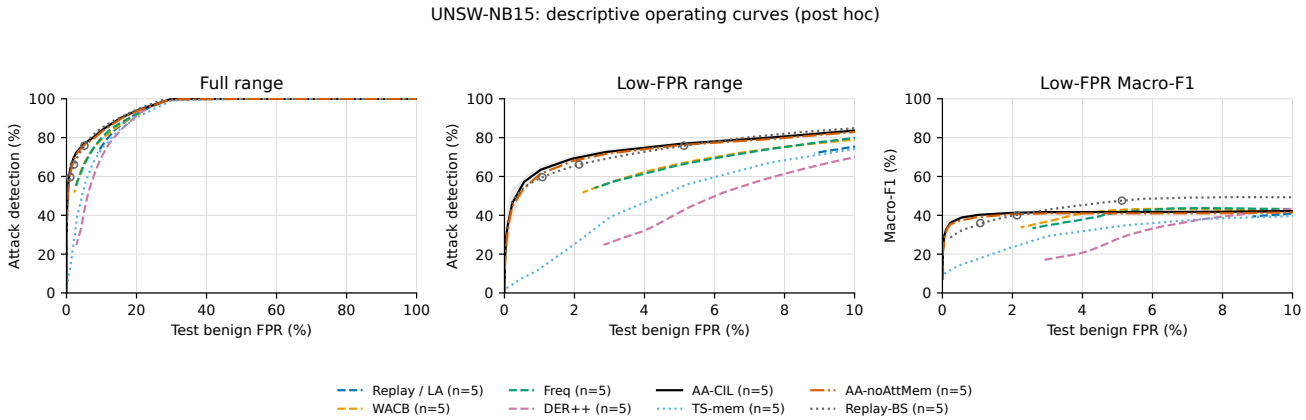



Figure 3: UNSW-NB15 operating curves (post hoc). Threshold-sweep summaries are descriptive, not calibration-set operating points or the registered test. Missing methods or seeds are identified in the generated panel.

mean final BAcc no more than two percentage points below original AA-CIL; and ADR not significantly below original AA-CIL. The two-point BAcc condition is a fixed mean-difference rule, not a significance or noninferiority test. CSE aggregate metrics remain unavailable here until complete five-seed groups and their guarded release exist; no CSE outcome may change the frozen choice. The pipeline additionally withholds its reference summaries until all 35 physical reference cells are complete. Missing external evidence does not support a resolved-trade-off claim.

# B Retention-extension paired comparisons

# C Fixed-KD reassessment

# D Fixed-capacity family counts

# E Additional calibration targets

These results reuse the permanently held-out calibration source. They are sensitivity results, not a search over test thresholds. Actual test FPR accompanies each target, and an undefined per-class quantity remains undefined.

# F Difficult-family performance

These selected families illustrate errors that binary attack detection alone cannot reveal. They are descriptive diagnostics, not a replacement endpoint or a selection rule for the main result; incomplete groups retain their actual seed counts.

# G Registered safety contrasts and timing components

The timing components retain the measurement boundaries of their source records. In particular, a CPU analysis attempt and its nested inference or threshold-search components must not be summed twice.

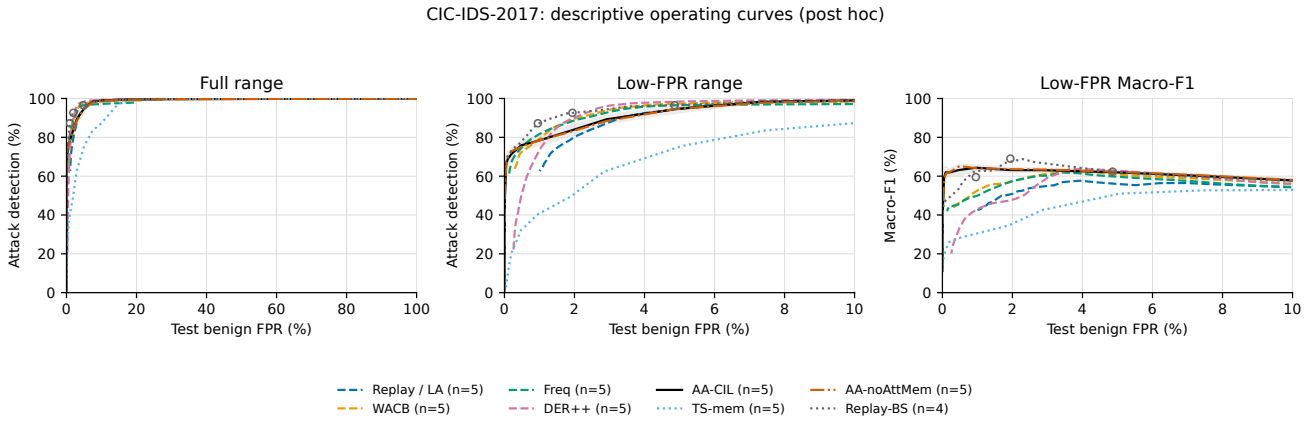



Figure 4: CIC-IDS-2017 operating curves (post hoc). Threshold-sweep summaries are descriptive, not calibration-set operating points or the registered test. Missing methods or seeds are identified in the generated panel.

Table 18: Descriptive final-task differences from the offline Joint reference at $\alpha = 2\%$. Values are AA-CIL minus Joint in percentage points; mean ± sample SD across paired completed seeds.

| Dataset | Paired n | Δ FPR | Δ detection | Δ MF1 |
|---|---|---|---|---|
| NSL | 5 | -0.641±1.101 | -3.795±4.125 | -3.706±5.262 |
| UNSW | 5 | -0.006±0.118 | 1.981±1.719 | 3.959±0.990 |
| CIC17 | 5 | 0.002±0.052 | -14.531±2.836 | -5.278±1.264 |

Joint uses offline old-training access and is neither a matched deployment baseline nor a guaranteed empirical upper bound. These gaps have no success threshold.

Table 19: Per-seed total training seconds across tasks and final memory occupancy. G and F are separately trained models; single-classifier methods have no G/F split. Timing excludes data loading/saving and is not an exclusive-GPU benchmark.

| Data | Method | n | G (s) | F (s) | Total (s) | Used rows | G/BS benign MiB |
|---|---|---|---|---|---|---|---|
| NSL | Replay | 5 | – | – | 31.0±0.4 | 852±0 | – |
| NSL | DER++ | 5 | – | – | 70.0±1.0 | 986±0 | – |
| NSL | WACB | 5 | – | – | 30.4±0.2 | 927±0 | – |
| NSL | Freq | 5 | – | – | 31.3±1.0 | 784±0 | – |
| NSL | AA-CIL | 5 | 35.2±1.4 | 32.0±1.4 | 67.1±2.7 | 802±0 | 22.51±0.00 |
| NSL | TS-mem | 5 | 31.7±1.0 | 31.3±1.3 | 63.0±2.0 | 852±0 | 0.08±0.00 |
| NSL | AA-noAttMem | 5 | 34.7±0.8 | 30.6±1.1 | 65.3±1.8 | 652±0 | 22.51±0.00 |
| NSL | Replay-BS (post hoc) | 5 | – | – | 31.5±3.0 | 802±0 | 22.51±0.00 |
| NSL | Joint (offline) | 5 | – | – | 45.7±0.8 | 86±0 | – |
| UNSW | Replay | 5 | – | – | 34.7±0.4 | 1927±0 | – |
| UNSW | DER++ | 5 | – | – | 80.0±1.6 | 1993±0 | – |
| UNSW | WACB | 5 | – | – | 34.2±1.1 | 2000±0 | – |
| UNSW | Freq | 5 | – | – | 33.6±0.7 | 1589±0 | – |
| UNSW | AA-CIL | 5 | 41.0±1.1 | 37.8±1.0 | 78.8±1.9 | 1905±0 | 13.35±0.00 |
| UNSW | TS-mem | 5 | 39.5±1.1 | 39.5±1.3 | 78.9±2.2 | 1927±0 | 0.05±0.00 |
| UNSW | AA-noAttMem | 5 | 40.5±0.6 | 37.9±0.4 | 78.4±0.9 | 1727±0 | 13.35±0.00 |
| UNSW | Replay-BS (post hoc) | 5 | – | – | 40.4±1.4 | 1905±0 | 13.35±0.00 |
| UNSW | Joint (offline) | 5 | – | – | 52.5±0.6 | 193±0 | – |
| CIC17 | Replay | 5 | – | – | 219.8±6.0 | 1430±0 | – |
| CIC17 | DER++ | 5 | – | – | 553.9±13.2 | 1585±0 | – |
| CIC17 | WACB | 5 | – | – | 223.0±3.8 | 1516±0 | – |
| CIC17 | Freq | 5 | – | – | 221.2±3.5 | 1452±0 | – |
| CIC17 | AA-CIL | 5 | 82.2±6.4 | 60.1±0.1 | 142.3±6.3 | 1402±0 | 15.07±0.00 |
| CIC17 | TS-mem | 5 | 54.0±4.2 | 53.3±3.5 | 107.4±7.7 | 1430±0 | 0.05±0.00 |
| CIC17 | AA-noAttMem | 5 | 70.9±2.8 | 53.0±1.9 | 123.9±4.3 | 1230±0 | 15.07±0.00 |
| CIC17 | Replay-BS (post hoc) | 4 | – | – | 69.4±4.2 | 1402±0 | 15.07±0.00 |
| CIC17 | Joint (offline) | 5 | – | – | 899.9±21.2 | 145±0 | – |

G/BS benign MiB reports final-task benign input feature-plus-index array bytes, not additional persistent storage. AA-CIL, AA-noAttMem and the single-model Replay-BS use up to 50,000 refreshed benign inputs outside M. TS-mem uses its retained training portion (normally 180 of 200 Normal slots) inside M; these bytes must not be added again to replay storage. Missing baseline byte records are shown as –, not zero. Dense-label materialization, permanent calibration features/indices, model weights and other transient copies are separate and excluded from this column; see the setup table. Joint is not a matched compute/storage baseline.

Table 20: Necessary comparisons under the validation-selected retention protocol.

| Data | Criterion | Reference | n | Δ (pp) | One-sided $p$ | Decision |
|---|---|---|---|---|---|---|
| UNSW | 1 | WACB | 5 | 11.84±2.91 | 0.00041 | fail |
| UNSW | 2 | Replay-BS (post hoc) | 5 | 16.67±1.26 | $3.868 \times 10^{-6}$ | pass |
| UNSW | 3 | AA-CIL | 5 | 1.92±1.29 | 0.98539 | pass |
| NSL | 1 | DER++ | 5 | 1.42±5.29 | 0.28991 | pass |
| NSL | 2 | Replay-BS (post hoc) | 5 | -0.33±2.73 | 0.59933 | fail |
| NSL | 3 | AA-CIL | 5 | 9.57±3.63 | 0.99794 | pass |

Candidate minus reference; exact five paired formal seeds. 1: all-old forgetting, $p_> < .05$ means harm/failure against the lowest-forgetting external baseline. 2: final BAcc, $p_> < .05$ is required against Replay-BS. 3: ADR, $p_< < .05$ means harm/failure against original AA-CIL. Only useful complete comparisons are shown; absent comparisons remain pending and are not passed. Both UNSW and NSL are required. No detected harm is not equivalence. Unadjusted tests.

Table 21: Completed descriptive AA-CIL-ISO minus AA-CIL-KD comparisons.

| Data | Metric | n | Δ (pp) | $p_<$ | $p_>$ | $p_{2s}$ |
|---|---|---|---|---|---|---|
| UNSW | Forget (all) | 5 | -2.29±1.03 | 0.00385 | 0.99615 | 0.00770 |
| UNSW | Mean inc. BAcc | 5 | -18.86±1.10 | $1.365 \times 10^{-6}$ | $1 - 1.365 \times 10^{-6}$ | $2.731 \times 10^{-6}$ |
| NSL | Forget (all) | 5 | -0.93±0.89 | 0.03981 | 0.96019 | 0.07963 |
| NSL | Mean inc. BAcc | 5 | -18.10±0.55 | $1.052 \times 10^{-7}$ | $1 - 1.052 \times 10^{-7}$ | $2.104 \times 10^{-7}$ |

Exact five paired formal seeds; mean±sample SD of differences in percentage points. These unadjusted comparisons neither select the main version nor add a success requirement. Undefined tests remain –; no seed is discarded. The original superiority and earlier retention decisions are unchanged.

Table 22: Fixed-KD reassessment at stagewise Normal-holdout calibration target $\alpha = 2\%$.

| Data | $n$ | Mean inc. BAcc | Final BAcc | Forget (pp) | ADR | MF1 | FPR |
|---|---|---|---|---|---|---|---|
| UNSW | 5 | 63.75±1.11 | 41.80±1.93 | 2.31±1.07 | 70.53±0.50 | 42.08±1.93 | 2.10±0.18 |
| NSL | 5 | 76.39±0.26 | 60.82±0.73 | 0.98±0.76 | 71.67±0.66 | 61.43±2.25 | 7.94±0.45 |
| CIC17 | 5 | 87.59±1.07 | 79.20±4.08 | -10.54±9.15 | 92.10±2.84 | 65.94±2.49 | 1.93±0.04 |

Mean±sample SD over completed trajectories at the stated $n$; $n = 1$ has undefined SD (–). Incomplete groups are descriptive only; significance requires all five formal seeds. UNSW is exploratory. NSL results were available before direct authorization, so NSL is not blind confirmation. KD uses fixed $\lambda_G = .5$ and DER $\alpha = \beta = 1$. The validation-selected CBF and original superiority decisions remain failed.

Table 23: Fixed-KD reassessment: candidate minus reference.

| Data | Criterion | Reference | $n$ | Δ (pp) | One-sided $p$ | Decision |
|---|---|---|---|---|---|---|
| UNSW | 1 | WACB | 5 | -1.17±3.41 | 0.7563 | pass |
| UNSW | 2 | Replay-BS (post hoc) | 5 | 3.78±1.43 | 0.002032 | pass |
| UNSW | 3 | AA-CIL | 5 | 1.44±0.85 | 0.9905 | pass |
| NSL | 1 | DER++ | 5 | 0.07±4.87 | 0.4876 | pass |
| NSL | 2 | Replay-BS (post hoc) | 5 | 1.49±2.58 | 0.1329 | fail |
| NSL | 3 | AA-CIL | 5 | 8.84±2.94 | 0.9987 | pass |
| CIC17 | 1 | WACB | 5 | -12.51±10.48 | 0.9721 | pass |
| CIC17 | 2 | Replay-BS (post hoc) | 4 | – | – | pending |
| CIC17 | 3 | AA-CIL | 5 | 8.44±2.61 | 0.999 | pass |

1: forgetting harm, $p_> < .05$ means failure against the lowest-forgetting external-five member. 2: final BAcc superiority over Replay-BS requires $p_> < .05$. 3: ADR harm relative to original AA means $p_< < .05$. NSL and CIC17 are both required; UNSW does not replace either. Unadjusted tests; no detected harm is not equivalence. Required numerical verdict: fail.

Table 24: Fixed family-model capacity of the selected per-candidate configurations.

| Data | Candidate | Training-resident | Optimizer | Inference | Extra components |
|---|---|---|---|---|---|
| UNSW | KD-weak | 26505 | 26505 | 26505 | 0 |
| NSL | KD-weak | 32004 | 32004 | 32004 | 0 |
| UNSW | KD+WA | 26505 | 26505 | 26505 | 0 |
| NSL | KD+WA | 32004 | 32004 | 32004 | 0 |
| UNSW | KD+CBF | 26505 | 26505 | 26505 | 0 |
| NSL | KD+CBF | 32004 | 32004 | 32004 | 0 |
| UNSW | EMA-KD | 53010 | 26505 | 26505 | 1 |
| NSL | EMA-KD | 64008 | 32004 | 32004 | 1 |
| UNSW | DUAL | 53001 | 53001 | 53001 | 1 |
| NSL | DUAL | 64004 | 64004 | 64004 | 1 |

F only; the common G and transient previous-task gate teacher are separate. Counts come from saved development architectures and remain fixed across tasks. EMA training-resident count includes its slow copy; only EMA is used at inference. DUAL uses two fixed encoders and one joint head within a single F.

Table 25: Additional calibration targets: NSL-KDD. Rates in percent; mean ± sample SD.

| Method | $\alpha$ | n | Test FPR | Detection | MF1 |
|---|---|---|---|---|---|
| Replay | 1 | 5 | 2.60±1.82 | 56.54±8.96 | 36.14±8.07 |
| Replay | 5 | 5 | 5.92±2.76 | 76.73±3.63 | 59.43±3.60 |
| DER++ | 1 | 5 | 1.45±0.35 | 58.01±6.83 | 42.04±7.13 |
| DER++ | 5 | 5 | 4.44±0.74 | 73.86±3.16 | 60.62±2.98 |
| WACB | 1 | 5 | 2.74±0.25 | 61.74±4.42 | 48.34±5.05 |
| WACB | 5 | 5 | 5.31±2.02 | 75.34±4.53 | 57.78±1.61 |
| Freq | 1 | 5 | 4.54±2.44 | 66.24±3.69 | 57.79±3.89 |
| Freq | 5 | 5 | 6.34±2.74 | 75.94±4.01 | 55.34±2.40 |
| Replay+LA | 1 | 5 | 2.60±1.82 | 56.54±8.96 | 36.14±8.07 |
| Replay+LA | 5 | 5 | 5.92±2.76 | 76.73±3.63 | 59.43±3.60 |
| TS-mem | 1 | 5 | 2.32±1.85 | 51.03±6.90 | 45.92±4.77 |
| TS-mem | 5 | 5 | 6.10±2.69 | 73.74±4.61 | 55.52±2.29 |
| AA-noAttMem | 1 | 5 | 3.59±1.57 | 57.20±1.49 | 56.68±3.42 |
| AA-noAttMem | 5 | 5 | 6.18±2.36 | 67.14±2.65 | 56.35±2.93 |
| AA-CIL | 1 | 5 | 4.89±1.91 | 58.93±2.19 | 53.54±2.62 |
| AA-CIL | 5 | 5 | 7.14±1.75 | 67.83±3.44 | 57.44±1.81 |
| Replay-BS (post hoc) | 1 | 5 | 3.16±1.58 | 59.01±4.04 | 58.43±2.41 |
| Replay-BS (post hoc) | 5 | 5 | 4.36±2.25 | 66.84±4.27 | 55.84±2.69 |
| Joint (offline) | 1 | 5 | 6.30±1.23 | 64.23±1.29 | 57.26±2.41 |
| Joint (offline) | 5 | 5 | 7.61±0.85 | 71.12±2.70 | 57.37±1.03 |

Table 26: Additional calibration targets: UNSW-NB15. Rates in percent; mean ± sample SD.

| Method | $\alpha$ | n | Test FPR | Detection | MF1 |
|---|---|---|---|---|---|
| Replay | 1 | 5 | 0.86±0.18 | 28.97±4.87 | 23.40±0.90 |
| Replay | 5 | 5 | 4.80±0.19 | 56.73±2.53 | 32.85±1.80 |
| DER++ | 1 | 5 | 0.99±0.14 | 9.69±5.65 | 11.23±2.31 |
| DER++ | 5 | 5 | 4.87±0.42 | 40.57±6.65 | 26.89±5.22 |
| WACB | 1 | 5 | 0.95±0.05 | 39.02±5.88 | 27.80±2.02 |
| WACB | 5 | 5 | 4.90±0.35 | 65.89±1.88 | 42.17±0.72 |
| Freq | 1 | 5 | 0.98±0.17 | 41.24±3.41 | 26.95±2.00 |
| Freq | 5 | 5 | 5.00±0.33 | 65.64±1.99 | 41.21±1.82 |
| Replay+LA | 1 | 5 | 0.86±0.18 | 28.97±4.87 | 23.40±0.90 |
| Replay+LA | 5 | 5 | 4.80±0.19 | 56.73±2.53 | 32.85±1.80 |
| TS-mem | 1 | 5 | 0.96±0.09 | 11.78±10.64 | 17.15±6.58 |
| TS-mem | 5 | 5 | 5.22±0.35 | 55.95±5.94 | 34.85±1.90 |
| AA-noAttMem | 1 | 5 | 1.02±0.12 | 61.27±3.58 | 39.15±0.80 |
| AA-noAttMem | 5 | 5 | 5.07±0.30 | 76.24±0.75 | 40.96±0.34 |
| AA-CIL | 1 | 5 | 1.04±0.09 | 63.51±1.40 | 40.22±0.39 |
| AA-CIL | 5 | 5 | 5.05±0.17 | 76.83±0.62 | 41.67±0.50 |
| Replay-BS (post hoc) | 1 | 5 | 1.08±0.12 | 59.65±0.93 | 36.02±1.56 |
| Replay-BS (post hoc) | 5 | 5 | 5.13±0.29 | 75.85±1.11 | 47.54±1.61 |
| Joint (offline) | 1 | 5 | 1.00±0.10 | 61.09±1.83 | 33.36±0.46 |
| Joint (offline) | 5 | 5 | 5.08±0.33 | 77.39±1.51 | 41.60±0.74 |

Table 27: Additional calibration targets: CIC-IDS-2017. Rates in percent; mean ± sample SD.

| Method | $\alpha$ | n | Test FPR | Detection | MF1 |
|---|---|---|---|---|---|
| Replay | 1 | 5 | 0.98±0.04 | 67.05±8.32 | 44.49±6.18 |
| Replay | 5 | 5 | 4.87±0.06 | 94.85±1.90 | 57.05±1.92 |
| DER++ | 1 | 5 | 0.96±0.03 | 72.99±7.31 | 42.92±2.54 |
| DER++ | 5 | 5 | 4.86±0.03 | 98.28±0.27 | 62.87±0.65 |
| WACB | 1 | 5 | 0.95±0.04 | 79.18±2.75 | 50.23±4.04 |
| WACB | 5 | 5 | 4.85±0.02 | 97.21±0.60 | 61.05±0.98 |
| Freq | 1 | 5 | 0.97±0.04 | 81.16±2.48 | 50.00±4.07 |
| Freq | 5 | 5 | 4.83±0.03 | 96.46±0.76 | 59.82±1.49 |
| Replay+LA | 1 | 5 | 0.98±0.04 | 67.05±8.32 | 44.49±6.18 |
| Replay+LA | 5 | 5 | 4.87±0.06 | 94.85±1.90 | 57.05±1.92 |
| TS-mem | 1 | 5 | 0.94±0.01 | 40.22±6.08 | 30.28±1.91 |
| TS-mem | 5 | 5 | 5.07±0.45 | 75.51±6.27 | 51.01±3.59 |
| AA-noAttMem | 1 | 5 | 0.98±0.01 | 78.32±1.40 | 64.10±1.21 |
| AA-noAttMem | 5 | 5 | 4.88±0.03 | 94.57±2.17 | 62.32±1.02 |
| AA-CIL | 1 | 5 | 0.96±0.01 | 78.23±2.71 | 64.35±1.36 |
| AA-CIL | 5 | 5 | 4.89±0.03 | 94.52±2.06 | 62.02±0.97 |
| Replay-BS (post hoc) | 1 | 4 | 0.95±0.03 | 87.25±1.74 | 59.51±6.41 |
| Replay-BS (post hoc) | 5 | 4 | 4.86±0.03 | 96.50±1.67 | 62.32±1.35 |
| Joint (offline) | 1 | 5 | 0.94±0.05 | 94.24±2.98 | 63.48±5.19 |
| Joint (offline) | 5 | 5 | 4.85±0.02 | 99.87±0.12 | 63.96±1.39 |

Table 28: Selected low-support or weak-family results at $\alpha = 2\%$, to expose failure modes hidden by binary detection. All classes remain in Macro-F1 and BAcc; only the attack guard excludes support below 20.

| Data | Method | Class | n | Support | $n_P$ | Precision | Recall | F1 |
|---|---|---|---|---|---|---|---|---|
| NSL | AA-CIL | R2L | 5 | 2885 | 5 | 79.21±20.34 | 19.34±5.56 | 30.98±8.50 |
| NSL | AA-CIL | U2R | 5 | 67 | 5 | 14.36±15.83 | 46.27±2.36 | 19.07±14.54 |
| NSL | TS-mem | R2L | 5 | 2885 | 5 | 49.71±22.71 | 9.74±4.42 | 16.03±6.95 |
| NSL | TS-mem | U2R | 5 | 67 | 5 | 6.68±2.86 | 40.30±8.31 | 11.24±4.03 |
| NSL | AA-noAttMem | R2L | 5 | 2885 | 5 | 79.36±13.06 | 15.45±4.49 | 25.71±6.83 |
| NSL | AA-noAttMem | U2R | 5 | 67 | 5 | 26.25±18.85 | 43.28±6.76 | 29.57±17.65 |
| NSL | Replay-BS (post hoc) | R2L | 5 | 2885 | 5 | 95.22±2.37 | 13.72±4.07 | 23.82±6.15 |
| NSL | Replay-BS (post hoc) | U2R | 5 | 67 | 5 | 25.86±22.55 | 47.76±8.24 | 27.30±16.01 |
| UNSW | AA-CIL | Fuzzers | 5 | 3750 | 5 | 69.58±4.05 | 9.67±1.14 | 16.94±1.67 |
| UNSW | AA-CIL | Worms | 5 | 32 | 5 | 3.52±0.45 | 63.75±6.09 | 6.68±0.83 |
| UNSW | TS-mem | Fuzzers | 5 | 3750 | 5 | 47.54±14.46 | 3.39±2.15 | 6.28±3.88 |
| UNSW | TS-mem | Worms | 5 | 32 | 5 | 7.62±3.64 | 21.88±15.78 | 9.00±1.30 |
| UNSW | AA-noAttMem | Fuzzers | 5 | 3750 | 5 | 67.67±4.11 | 9.11±0.37 | 16.06±0.67 |
| UNSW | AA-noAttMem | Worms | 5 | 32 | 5 | 3.73±0.57 | 64.38±8.44 | 7.06±1.06 |
| UNSW | Replay-BS (post hoc) | Fuzzers | 5 | 3750 | 5 | 68.07±5.79 | 12.20±3.50 | 20.45±4.99 |
| UNSW | Replay-BS (post hoc) | Worms | 5 | 32 | 1 | 100.00 | 0.62±1.40 | 1.21±2.71 |
| CIC17 | AA-CIL | Botnet | 5 | 370 | 5 | 6.75±0.96 | 95.41±1.74 | 12.59±1.66 |
| CIC17 | AA-CIL | Infiltration | 5 | 6 | 5 | 2.17±1.25 | 66.67±0.00 | 4.16±2.34 |
| CIC17 | TS-mem | Botnet | 5 | 370 | 5 | 0.72±0.29 | 5.73±2.00 | 1.25±0.48 |
| CIC17 | TS-mem | Infiltration | 5 | 6 | 5 | 0.22±0.08 | 63.33±7.45 | 0.44±0.15 |
| CIC17 | AA-noAttMem | Botnet | 5 | 370 | 5 | 6.50±1.01 | 96.05±2.38 | 12.15±1.74 |
| CIC17 | AA-noAttMem | Infiltration | 5 | 6 | 5 | 3.36±2.00 | 66.67±0.00 | 6.31±3.53 |
| CIC17 | Replay-BS (post hoc) | Botnet | 4 | 370 | 4 | 30.35±7.48 | 60.68±0.84 | 40.03±6.60 |
| CIC17 | Replay-BS (post hoc) | Infiltration | 4 | 6 | 4 | 9.21±6.83 | 41.67±9.62 | 14.17±9.42 |

Precision uses defined seeds only; $n_P$ is defined_n and n is the full completed seed count. Original null precision values remain null in the generated JSON. With no defined value the mean is –; with fewer than two the sample SD is undefined. No seed is removed from other defined endpoints. Complete per-class confusion matrices accompany the saved result JSON.

Table 29: Remaining registered contrasts. Differences are AA-CIL minus the named comparator. The harm tests use the less-than alternative; the CIC gate-refresh ablation uses the greater-than alternative.

| Data | Endpoint | Comparator | Δ (pp) | $p$ | Outcome |
|---|---|---|---|---|---|
| NSL | MF1 vs strongest | WACB | -1.365±5.922 | 0.31675 | No detected harm |
| NSL | Guard vs Replay | Replay | 14.534±7.440 | 0.99401 | No detected harm |
| UNSW | MF1 vs strongest | AA-noAttMem | 0.746±0.638 | 0.97037 | No detected harm |
| UNSW | Guard vs Replay | Replay | 33.232±0.851 | 1.00000 | No detected harm |
| CIC17 | MF1 vs strongest | AA-noAttMem | -0.455±1.756 | 0.29686 | No detected harm |
| CIC17 | Guard vs Replay | Replay | 41.399±6.098 | 0.99995 | No detected harm |
| CIC17 | MF1 gate-refresh ablation | TS-mem | 28.522±1.744 | 0.00000 | PASS |

Only complete five-seed groups determine these outcomes. No detected harm does not establish equivalence or noninferiority. Criterion 5 is descriptive completion of Joint-gap and cost reporting, not superiority over Joint.

Table 30: Per-seed recorded overhead across tasks (seconds, mean ± sample standard deviation). Separate post-training CPU analyses are shown without adding their nested subcomponents.

| Data | Method | n | Selection | In-run calib. | In-run eval. | CPU curves | CPU ext. calib. |
|---|---|---|---|---|---|---|---|
| NSL | Replay | 5 | 0.022±0.013 | – | 0.113±0.011 | 0.112±0.014 | 0.132±0.008 |
| NSL | DER++ | 5 | 0.024±0.002 | – | 0.068±0.005 | 0.136±0.011 | 0.135±0.008 |
| NSL | WACB | 5 | 0.016±0.001 | – | 0.058±0.004 | 0.119±0.010 | 0.143±0.007 |
| NSL | Freq | 5 | 0.017±0.001 | – | 0.064±0.006 | 0.128±0.007 | 0.146±0.005 |
| NSL | AA-CIL | 5 | 0.082±0.003 | 0.025±0.010 | 0.081±0.013 | 0.058±0.004 | – |
| NSL | TS-mem | 5 | 0.027±0.001 | 0.020±0.003 | 0.079±0.006 | 0.063±0.017 | – |
| NSL | AA-noAttMem | 5 | 0.081±0.001 | 0.020±0.002 | 0.080±0.002 | 0.058±0.005 | – |
| NSL | Replay-BS (post hoc) | 5 | 0.083±0.004 | 0.325±0.016 | 0.058±0.004 | 0.110±0.028 | – |
| NSL | Joint (offline) | 5 | 0.012±0.001 | – | 0.059±0.003 | 0.149±0.019 | 0.154±0.007 |
| UNSW | Replay | 5 | 0.029±0.002 | – | 0.172±0.004 | 0.158±0.018 | 0.217±0.020 |
| UNSW | DER++ | 5 | 0.036±0.003 | – | 0.085±0.008 | 0.197±0.017 | 0.236±0.022 |
| UNSW | WACB | 5 | 0.029±0.002 | – | 0.085±0.005 | 0.196±0.010 | 0.210±0.016 |
| UNSW | Freq | 5 | 0.027±0.001 | – | 0.080±0.004 | 0.184±0.009 | 0.197±0.015 |
| UNSW | AA-CIL | 5 | 0.074±0.006 | 0.022±0.001 | 0.118±0.007 | 0.061±0.004 | – |
| UNSW | TS-mem | 5 | 0.024±0.003 | 0.027±0.006 | 0.124±0.006 | 0.066±0.014 | – |
| UNSW | AA-noAttMem | 5 | 0.077±0.009 | 0.023±0.002 | 0.117±0.008 | 0.072±0.027 | – |
| UNSW | Replay-BS (post hoc) | 5 | 0.070±0.005 | 0.540±0.040 | 0.100±0.008 | 0.190±0.032 | – |
| UNSW | Joint (offline) | 5 | 0.022±0.002 | – | 0.087±0.003 | 0.228±0.006 | 0.213±0.019 |
| CIC17 | Replay | 5 | 0.151±0.012 | – | 2.233±0.063 | 3.239±0.269 | 3.038±0.153 |
| CIC17 | DER++ | 5 | 0.187±0.021 | – | 1.246±0.099 | 3.674±0.131 | 3.257±0.176 |
| CIC17 | WACB | 5 | 0.147±0.010 | – | 1.203±0.093 | 3.470±0.180 | 3.135±0.094 |
| CIC17 | Freq | 5 | 0.149±0.006 | – | 1.204±0.068 | 3.623±0.061 | 3.170±0.173 |
| CIC17 | AA-CIL | 5 | 0.291±0.018 | 0.374±0.126 | 1.919±0.061 | 1.216±0.081 | – |
| CIC17 | TS-mem | 5 | 0.248±0.027 | 0.346±0.034 | 1.648±0.155 | 1.022±0.076 | – |
| CIC17 | AA-noAttMem | 5 | 0.286±0.022 | 0.324±0.020 | 1.649±0.096 | 1.169±0.099 | – |
| CIC17 | Replay-BS (post hoc) | 4 | 0.268±0.018 | 5.444±0.280 | 1.579±0.079 | 2.274±0.032 | – |
| CIC17 | Joint (offline) | 5 | 0.032±0.001 | – | 1.198±0.023 | 3.672±0.124 | 3.216±0.149 |

Selection, in-run calibration and in-run evaluation use the recorded buffer-selection, calibration and evaluation totals respectively. A missing key is –, never an assumed zero. AA-family and Replay-BS CPU curves use the source-matched recorded cell-total seconds from the independent post-hoc curve ledger when available; missing or mismatched cell records remain –. For older baselines the separate CPU curve-analysis and extended-calibration elapsed totals are not embedded in the in-run columns; nested holdout/search/inference sub-timings must not be added again. Shared resume/render-attempt overhead remains in the independent curve ledger and is excluded here. These columns are not a single end-to-end walltime total and do not duplicate Replay+LA, which shares Replay weights and calibration.

# References


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